

Unit 1 - Problem Solving with AI

▼ Teacher resources

▼ Printing Options

As AI becomes increasingly integrated into daily life, students need to move beyond simply using AI tools—they must develop the skills to critically evaluate, refine, and apply AI-generated content in responsible ways. This unit introduces students to AI as a collaborator in analytical and creative work, helping them explore where AI succeeds, where it fails, and how human oversight improves AI-driven problem solving.

This unit blends hands-on AI experimentation with critical evaluation and refinement, guiding students through an iterative cycle of testing, analyzing, and improving AI-generated outputs. Instead of passively accepting AI responses, students will actively investigate AI's reasoning, challenge its assumptions, and refine its outputs through computational thinking and ethical frameworks.

► 🎯 By the end of this unit, students will be able to . . .

Finished Teaching This Unit?

Answer this [short survey](#) to let the Code.org curriculum team know how the unit went.

Week 1	Lesson 1: Talking to Machines	Lesson 2: Beyond Words	Lesson 3: The AI's Brain	Lesson 4: Smart or Just Predictable?	Lesson 5: Uncovering Contradictions
Week 2	Lesson 6: Understanding Bias	Lesson 7: AI's Wild Imagination	Lesson 8: Debugging and Refining Outputs	Lesson 9: AI as a Co-Creator	Lesson 10: Remixing, Creativity, and Originality
Week 3	Lesson 11: Ownership, Ethics, and Creativity	Lesson 12: AI's Role in Society	Lesson 13: AI Time Capsule Project		

Key	☐ Instructional Lesson ☒ Assessment ☒ Unplugged Lesson
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▼ Pre-Assessment

▼ Pre-Assessment

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1-3

Pre-Assessment



▼ Content

▼ Lesson 1: Talking to Machines

How can I use AI effectively?

In this lesson, students explore how AI generates responses by recognizing patterns and predicting likely outcomes. Through a “Yes, and...” game, they simulate AI’s response-building process. They experiment using text and images to craft effective prompts using AI Chat levels. By the end, they understand how AI processes information and how to use strategic prompting for better results.



1

AI Predictions



2-4

Prompting AI

2

3

4



5

Prompting Practice



6



Check Your Understanding

▼ Lesson 2: Beyond Words

How does AI process different types of media?

In this lesson, students explore how multimodal AI models process and interpret different types of media by comparing AI-generated responses to text-only vs text and image inputs. Through hands-on activities and discussions, they analyze AI’s capabilities, limitations, and ethical considerations, reinforcing their understanding of effective prompting and privacy risks in AI applications.



1-2

AI's Image Observation Skills

1

2



3

Scenario-Based Prompting



4



Check Your Understanding

▼ Lesson 3: The AI's Brain

Career Journeys

How do AI models learn?

In this lesson, students uncover how large language models (LLMs) are trained, how they use neural networks to predict text, and how context shapes AI responses. Through an unplugged neural network activity, students simulate how AI generates predictions with and without context. They explore how LLMs rely on pattern recognition, probability, and human feedback to improve over time. Finally, students experiment with zero-shot, one-shot, and multi-shot prompting to see how different levels of input impact AI's output.

1-4

Prompting Challenges

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▼ Lesson 4: Smart or Just Predictable?

Career Journeys

Where does AI excel vs struggle?

Students explore the strengths and weaknesses of AI by comparing AI-generated responses with human responses for different tasks. They learn what AI hallucinations are and how to be critical of AI-generated output. Through hands-on activities, discussions, and real-world applications, students assess AI's reliability and determine when human oversight is necessary. By the end of the lesson, students justify when AI should or shouldn't be used in various professional contexts, developing a critical mindset toward AI's role as a collaborator.

1-4

AI vs Human Tasks

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Check Your Understanding

▼ Lesson 5: Uncovering Contradictions

Career Journeys

Can AI contradict itself?

Students will be asked to explore the reasons why AI might provide contradictory answers. They will practice refining prompts in order to both induce and eliminate contradictions in AI-generated responses. Additionally, they will examine ways in which AI contradictions can have real-world effects.



1

Two Truths and a Lie



2

Examining AI Contradictions Activity Guide



3-4

AI Contradictions in Practice

3

4



5-6

Try Your Own

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6



7-8

Check Your Understanding

7

8

▼ Lesson 6: Understanding Bias

Career Journeys

What is AI bias, and how does it emerge?

In this lesson, students explore how AI bias emerges from biased training data. Students analyze results from online search engines and AI Chat to examine them for bias. In the main activity, students select candidates for a particular job by filtering resumes. They compare their selections with those made by AI and note any bias in the selection process. Throughout the lesson, students brainstorm and discuss strategies to make AI systems more fair and ethical.



1

Career Clothing



2

Check Your Understanding



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Decision-Making Roles



4-5

AI Audit Challenge

4

5

▼ Lesson 7: AI's Wild Imagination

What are some examples of AI hallucinations? What are their impacts?

This lesson explores the real-world impacts of AI hallucinations. Building on their previous understanding of AI errors, students analyze real-world case studies where AI misinformation has caused harm. Through discussion, case studies, and fact-checking strategies, students evaluate AI's reliability and consider its ethical implications in fields like law, healthcare, and journalism.



Reduce AI Hallucinations



Check Your Understanding

▼ Lesson 8: Debugging and Refining Outputs

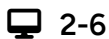
Career Journeys

Can we improve AI's responses?

Students explore how to improve AI-generated responses by refining their prompts using a five-step framework: task, context, reference, format, and tone. Through hands-on exploration in AI Chat levels, they test different prompt modifications, analyze how AI responds to varying levels of detail, and apply debugging techniques to enhance accuracy. At the end of the lesson, students summarize their learning about AI prompt refinement by adding ideas to a classroom Anchor Chart.



Adding a Task



AI Prompting Framework

2

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6



Check Your Understanding

7

8

9

▼ Lesson 9: AI as a Co-Creator

Career Journeys

How can AI enhance creativity?

In this lesson, students take on creative roles to design and launch a product or app that helps students succeed in school. Working in teams, they collaborate with AI across four stages—research, brainstorming, content generation, and editing—to co-create campaign materials like slogans, scripts, and visuals. The activity emphasizes decomposition, prompting strategy, and human-AI collaboration, helping students develop both technical and creative problem-solving skills.



Collaborate on a Product Launch

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▼ Lesson 10: Remixing, Creativity, and Originality

Career Journeys

How do human strengths and AI capabilities interact?

This lesson explores how AI can support human creativity through a hands-on remixing challenge. Students interact with AI to refine both their own creative content and AI-generated work, comparing the role of human input in both processes. The lesson emphasizes that while AI can enhance creative work, human judgment is essential for originality, nuance, and meaning.

 1

AI in Music Lab

 2

Students as the Pilot

 3-4

Students as the Co-Pilot

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 5

 Check Your Understanding

▼ Lesson 11: Ownership, Ethics, and Creativity

What are the ethical considerations of AI collaboration?

Students explore the ethical tradeoffs of using AI in creative contexts by engaging with real-world controversies involving ownership, attribution, and compensation. Through a values-based warm-up, rotating discussion stations, and a reflection on guiding principles, students consider how AI-generated content affects creators and what fairness looks like in these scenarios. The lesson helps students evaluate AI's role as a collaborator and justify when its use may or may not be appropriate.

 1

Ownership, Ethics, and Creativity

 2

 Check Your Understanding

▼ Lesson 12: AI's Role in Society

How should AI be used in solving real-world problems?

Students choose a real-world example of AI in use within the community, investigating who might benefit and who might be harmed from its implementation. They use their research to draft a call to action in which they propose solutions to the perceived issues that they uncover.



1

Articles



2

AI in the Community



3

Check Your Understanding



4

Optional: Collaborate with AI

▼ End of Unit Project

▼ Lesson 13: AI Time Capsule Project

In this project, students work with AI to develop a future prediction and create a presentation that explains their process. They collaborate with AI to generate and refine ideas, evaluate outputs for accuracy and bias, and reflect on how they shaped the final prediction. The project centers on ethical reasoning, student voice, and thoughtful use of multimodal inputs to support transparency and agency in AI collaboration.



1

AI Prediction



2

Final Product



3

End of Unit Survey

▼ End of Unit Assessment and Survey

▼ End of Unit Assessment

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1-3

End of Unit Assessment



▼ End of Unit Survey

☰ 1-3

End of Unit Survey

- 1 ✓
- 2 ✓
- 3 ✓

Lesson 1: Talking to Machines

45 minutes

Overview

How can I use AI effectively?

In this lesson, students explore how AI generates responses by recognizing patterns and predicting likely outcomes. Through a "Yes, and..." game, they simulate AI's response-building process. They experiment using text and images to craft effective prompts using AI Chat levels. By the end, they understand how AI processes information and how to use strategic prompting for better results.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- ▶ **AP** - Algorithms & Programming
- ▶ **CS** - Computing Systems

Agenda

Warm Up (10 minutes)

Yes, and...

Activity (30 minutes)

Problem Solving with AI Video

AI Predictions

Prompting AI

Wrap Up (5 minutes)

Rapid Roundtable

Objectives

Students will be able to:

- Analyze patterns in AI-generated responses to explain how input prompts influence outputs.
- Experiment with different prompts to observe how AI's responses change.
- Explain that AI models use probability and statistics to generate responses.
- Refine AI-generated outputs by iterating on prompts and recognizing patterns in responses.

Preparation

- Print copies of the Problem Solving with AI Unit Guide (one for each student)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **[KEY] Problem Solving with AI Unit Guide Exemplar** - Answer Key
▼ Make a Copy

For the students

- **Plan a Perfect Trip with ChatGPT** - Video
- **Problem Solving with AI** - Video ([Download](#))
- **Problem Solving with AI Unit Guide** - Activity Guide
▼ Make a Copy

Vocabulary

- **abstraction** - focusing on the important information and ignoring irrelevant details
- **artificial intelligence (AI)** - a technology that mimics human intelligence, performing tasks such as understanding language, recognizing patterns, and making decisions
- **probability** - the likelihood that a specific outcome might occur
- **prompt** - a question, instruction, scenario, or statement provided by the user to guide the AI's response

Teaching Guide

Warm Up (10 minutes)

Yes, and...

☒ **Do This:** Group students in pairs or groups of three and read aloud the sentence on the board. Have students take turns one-by-one building on the conversation by starting their next sentence with "Yes, and.."

Teaching Tip

For example:

- **Opening Statement:** "A mysterious talking cat started showing up to math class."
- **Student 1:** "Yes, and he said his name is Whiskers."
- **Student 2:** "Yes, and he happens to be very good at multiplication."
- **Student 1:** "Yes, and he has a sister who started showing up to English class."

(Optional) If you have time to play multiple rounds of the game, here are some additional opening statements:

- "The class down the hall just discovered a hidden tunnel beneath the school."
- "The principal announced that prom will be held on a spaceship this year."
- "Every locker in the main hallway suddenly started playing music."

 **Discuss:** *Summarize the story you came up with. What were the last two sentences of your conversation?*

- Have several groups summarize where their conversation went. Then have them share out their last sentence or two that they came up with.

 **Discussion Goal:** Students should recognize that all groups started with the same sentence starter but may have ended up with very different conversations. Have students note any patterns between the stories as well as any differences. This game leads into the main activity by explaining how AI generates content.

☒ **Do This:** Have students read the different sentences on this slide.

Remarks

Generative AI is like a “Yes, and” machine. It builds on existing sentences by finding a word or phrase that would make sense to come next.

☒ **Do This:** Introduce the lesson objectives.

☒ **Do This:** Introduce the Question of the Day.

Activity (30 minutes)

Problem Solving with AI Video (3 minutes)

 **Video:** Show Problem Solving with AI video.

Teaching Tip

Teaching Tip To encourage active engagement and reflection, use one or more of the strategies discussed in the [Guide to Curriculum Videos](#).

AI Predictions (7 minutes)

 **Distribute:** Give each student a copy of the Problem Solving with AI Unit Guide.

☒ **Do This:** Define *artificial intelligence*.

 **Discuss:** *Predict the word that you think will probably come next in the following sentence: The dog barked at the...*

- Ask students to predict the word that they think will probably come next (i.e. the most common word) and provide a reason for why they chose that word. Have all students raise their hand when they think of an answer. Start calling on students and have other students put down their hand if someone is called on who has the same answer.

 **Discussion Goal:** Students will see that there are several common possibilities that are chosen. Common words might include squirrel, mailman, stranger, etc. However, there will also be outliers that might not be as common (e.g. my aunt Lucy). Having students recognize that there are more and less common responses is a first step towards understanding how probability works while making this sort of prediction.

☒ **Do This:** Model the AI Chat in Lesson 1, Level 1 for students.

 1

AI Predictions

 **Transition:** Have students experiment with using the AI Chat to complete the sentence.

Teaching Tip

Have AI generate several outcomes to complete the sentence. You can keep prompting AI Chat using the same prompt or follow up by prompting it to "generate three more answers". You may also want to have students try out the prompt on their devices and compare responses.

 **Discuss:** *How does AI's output compare to the word you predicted?*

💡 **Discussion Goal:** Students should start to see how AI selects words based on probability. It tends to choose the most likely next word, which often mirrors our own prediction of the most-likely next word.

☑ **Do This:** Define *probability*.

☑ **Do This:** Read through this scenario of probability-driven predictions that make sense and are successful. Make sure students understand that the grey box represents the user input, and the blue box below represents the AI Chat response.

🎤 *Remarks*

AI uses the computational skill of pattern recognition to generate its most likely response. For example, it reads over thousands of sentences that contain the different phrases above (e.g. "heard the knock at the...") and starts to recognize a pattern.

☑ **Do This:** Examine this scenario where probability-driven predictions fail because they don't rely on meaning.

☑ **Do This:** Explain that AI focused on the words "key", "on", and "table" and abstracted away unnecessary details of the prompt.

🎤 *Remarks*

Since AI doesn't understand meaning, it drew a single key resting on the table in an unrealistic way. It prioritized the literal meaning of the essential details of the sentence rather than considering the sentence as a whole.

☑ **Do This:** Define *abstraction*.

Prompting AI (20 minutes)

☑ **Do This:** Define *prompt*.

💻 **Transition:** Direct students to Lesson 1, Level 2 to complete Levels 2 through 4.

- **Level 2:** Students input "Give me 3 gift ideas" into AI Chat.
- **Level 3:** Students input a more specific prompt that defines the person who is getting the gift.
- **Level 4:** Students add an image to accompany the specific prompt that they reuse from Level 3.



2-4

Prompting AI

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🎥 **Video:** Show the "Plan a Perfect Trip with ChatGPT" video.

Questions to consider with the video: What tips might you give to someone else who is looking to improve their prompts?

- What tips might you give to someone else who is looking to improve their prompts?

💡 **Discussion Goal:** Students might point out that they should include both information that they are interested in getting, but also information that they are not. They might also mention that including demographic information and specific numbers are helpful for more useful responses.

💡 Teaching Tip

Videos are used throughout the curriculum to spark discussions, supplement key concepts with additional explanations and examples, and expose students to the various roles and backgrounds of individuals in computer science.

While interacting with the video, turn on closed captioning so students can also read along as they watch.

To encourage active engagement and reflection, use one or more of the strategies discussed in the

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While interacting with the video, turn on closed captioning so students can also read along as they watch.

To encourage active engagement and reflection, use one or more of the strategies discussed in the [Guide to Curriculum Videos](#).

 **Transition:** Direct students to Lesson 1, Level 5. Students choose one of the AI Chat levels that is configured for a specific task. They start by giving the AI Chat a generic prompt first (such as "Help me prepare for a class presentation."), then make it more specific, then make it more specific and add an image. Students should update their prompt or interact with the AI Chat in at least three back and forth interactions.

5

Prompting Practice

Teaching Tip

This is a choice level, so students only need to complete one of the levels. However, if students finish early or are looking for an additional challenge, they can choose Option D: Debate an AI Partner level. This option is more challenging because it requires students to engage in a back-and-forth exchange with AI, refine their arguments, and adapt their reasoning based on AI's responses. Unlike other choices that focus on planning tasks, this level encourages students to critically analyze AI-generated arguments and practice constructing persuasive responses.

☒ **Do This:** Direct students to Lesson 1, Level 6 to complete the formative assessment. Display students' anonymous responses.

6

Check Your Understanding

Teaching Tip

This level can be used as a formative assessment to check student understanding of the lesson objectives. Here are things to look for and actions you can take:

- If students are only describing what happened rather than why it happened, prompt them to revisit earlier lesson segments on probability-driven outputs. Consider a quick group discussion where students share examples of prompts that worked well and why those prompts influenced AI's response.
- If students seem uncertain about constructing high-quality prompts, have them re-examine Level 5 where best prompting strategies are summarized. They can also do a brief pair activity—each student provides a vague prompt, then a specific prompt, and compares the results.
- If many students need more help, provide short demos or additional videos showing how multimodal inputs can change the AI's outputs. Or point them to the optional video about planning a family trip to emphasize how refining prompts improves results. Encourage a follow-up activity where students generate multiple answers to the same prompt, then analyze and explain what changed.

Wrap Up (5 minutes)

Rapid Roundtable

☒ **Do This:** Review the lesson objectives.

☒ **Do This:** Review the Question of the Day.

☒ **Do This:** Have students sit in a circle or small groups. Pose the Question of the Day to students. One by one, they quickly state their answer to the Question of the Day. Responses should be brief (one sentence or key phrase). They should see how many unique answers they can come up with.

Teaching Tip

If students are struggling to answer the Question of the Day, have them think about which prompting strategies got them the most helpful AI Chat outputs. For example, ask students what details they included in their prompts or how they selected an image to add to their prompt.

☒ **Do This:** Review the Key Vocabulary.



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Lesson 2: Beyond Words

45 minutes

Overview

How does AI process different types of media?

In this lesson, students explore how multimodal AI models process and interpret different types of media by comparing AI-generated responses to text-only vs text and image inputs. Through hands-on activities and discussions, they analyze AI's capabilities, limitations, and ethical considerations, reinforcing their understanding of effective prompting and privacy risks in AI applications.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- ▶ **AP** - Algorithms & Programming
- ▶ **IC** - Impacts of Computing

Agenda

Warm Up (10 minutes)

Activity (30 minutes)

Add Images

Scenario-Based Multimodal Prompting

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Compare how AI-generated outputs change based on the type of input provided.
- Experiment with different media inputs to observe AI's interpretation and limitations.
- Explain that multimodal AI models process information from multiple types of input.
- Identify privacy and ethical concerns when using AI with multimedia data.

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **[KEY] Problem Solving with AI Unit Guide Exemplar** - Answer Key
▼ Make a Copy

For the students

- **Problem Solving with AI Unit Guide** - Activity Guide
▼ Make a Copy

Vocabulary

- **multimodal model** - an AI system that can process information from multiple types of input

Teaching Guide

Warm Up (10 minutes)

 **Discuss:** Click through the animated slide to display the prompts:

- *If a friend wanted to locate a certain bird in the wild, how would you help them? Would you give them . . .*
- *a written description?*
- *a picture?*
- *a video?*
- *an audio recording of its call?*
- *something else?*
- *Why?*

 **Discussion Goal:** Students should recognize that different types of media serve different purposes. Text provides details, images help with identification, and audio and video capture sound and movement. AI must process different inputs to generate useful responses.

Teaching Tip

If students struggle to pick a media type, prompt them with follow up questions, such as: "How do YOU recognize a bird in the wild?" or "Would text be enough to help a friend who has never seen this bird before?"

Remarks

AI models like GPT-4, that powers ChatGPT, are trained to analyze multiple types of input (text, images, audio). Some AI models input text only, some input text and images, some input video, etc. Our AI Chat allows us to input text, images, and PDFs.

☒ **Do This:** Define *multimodal model*. Provide students with some real-world examples, such as Google Lens, self-driving cars, ChatGPT's image recognition, etc.

Teaching Tip

If students ask about AI that uses all four types of input (text, image, video, and sound), acknowledge that some AI models specialize in one type while others combine them. Self-driving cars, for example, use vision and audio sensors.

Remarks

Multimodal AI isn't just used in chatbots – it's changing industries.

☒ **Do This:** Use the examples on the slide to show how AI uses multiple inputs in real-world applications.

☒ **Do This:** Introduce the lesson objectives.

☒ **Do This:** Introduce the Question of the Day.

Activity (30 minutes)

Add Images (10 minutes)

☑ **Do This:** Have students take out their **Problem Solving with AI Unit Guide**.

☑ **Do This:** Have students list everything they observe in the image found in the "What's in an image?" section of Lesson 2 in their Unit Guides.

🔄 **Circulate:** As students are listing observations, listen for students focusing only on objects rather than details like lighting, time of day, or social context. If needed, ask: "What details beyond objects help describe this image?"

Remarks

Now that you have noted what you see in the image, let's see how AI describes it.

🖥️ **Transition:** Direct students to Lesson 2, Level 1 to complete Levels 1 and 2.

- **Level #1:** Students review multimodal concepts
- **Level #2:** Students add an image and ask AI Chat to describe it. Students should complete the "What's in an image?" and "Privacy and Ethical Concerns" sections in their Unit Guide.

🖥️ 1-2

AI's Image Observation Skills

1

2

🔄 **Circulate:** Look for students noticing unexpected AI-detected details (e.g., time of day, inferred location). If students struggle, ask students, "Did AI describe something you missed?" or "Did AI infer anything beyond what's in the image?"

Teaching Tip

If students struggle to compare their observations with AI's response, ask: "Did AI notice something you didn't? Why might that be?"

🗣️ **Discuss:** Click through the animated slide to display the prompts:

- *What should we consider before sharing images online with AI tools?*
- *What are the benefits and risks of AI interpreting images without human context?*

💡 **Discussion Goal:** Students should understand that AI can infer more than just visible objects in an image, including time, location, and personal details. They should critically evaluate privacy risks and ethical concerns when sharing images with AI.

Teaching Tip

For responses related to ethical considerations - If students give general answers, push for specificity by asking: "What's one real-world example of when an AI-generated image could cause harm?"

For responses related to privacy concerns - If students don't connect privacy risks to real-world consequences, ask: "How could an AI misinterpret an image and make incorrect assumptions?"

If time allows, prompt the class with: "Based on what was shared, do you think AI should be allowed to process certain types of images? Why or why not?" to get students thinking about ethics of AI tool use.

☒ **Do This:** Share this example of AI models increasingly having reverse image search capabilities. Providing an image of a location into newer AI models will often output a very accurate guess as to the image's location using only context clues present in the image.

Teaching Tip

This YouTube Short (<https://youtube.com/shorts/qClzSxOQGVI?si=oKNSi7LIAPgQNF1p>) shows how a human uses context clues from an image to find the location within a music video. It illustrates some of the ways in which context is used to pinpoint locations. This article from Tech Crunch also details the debate about privacy and ethics that is arising from this new AI ability: <https://techcrunch.com/2025/04/17/the-latest-viral-chatgpt-trend-is-doing-reverse-location-search-from-photos/>

Scenario-Based Multimodal Prompting (20 minutes)

 **Transition:** Direct students to Lesson 2, Level 3. Students prompt the chatbot to suggest improvements, first based off of a text-only description and then based on an image.

 3

Scenario-Based Prompting

 **Circulate:** Watch for students who struggle to explain why AI's response changed when using text vs. images. If needed, ask: "What new details did AI include when you used an image? Was the response more specific or more general?"

☒ **Do This:** Direct students to Lesson 2, Level 4 to complete the formative assessment.

 4

Check Your Understanding

☒ **Do This:** Display students' anonymous responses, and choose one of the three approaches below based on how students did:

- **Option 1: Whole-Class Poll & Debrief (Quick Check-In)**
 - Display the poll results for the multiple choice question.
 - Call on 2-3 students from each answer group (Text Only vs Text +Image) to explain their reasoning
 - Wrap up by asking students: "How does this confirm or challenge what we learned about effective prompting?"
- **Option 2: Small Group Discussion & Class Share-Out (Collaborative)**
 - Assign students to groups of 3-4.
 - Each group discusses: How did AI's response change when you included an image? and What do you think influenced AI's different responses?
 - Groups write down 1-2 insights on a shared board (physical or digital).
 - Teacher facilitates a brief whole-class discussion based on key themes.
- **Option 3: Silent Gallery Walk (Reflective)**
 - Post anonymized student responses using the "View Student Responses" Pin Response feature.
 - Students independently respond to pinned responses using:

- Agree/Disagree: Do they agree with the reasoning? Why?
- Question: What would they ask this student to clarify?
- Connection: How does this relate to prior learning?
- End with whole-class synthesis: What patterns or surprises did students notice?

✓ Assessment Opportunity ▲

This level can be used as a formative assessment to check student understanding of the lesson objectives. Here are things to look for and actions you can take:

- If most students correctly identified how AI's response changed with an image, reinforce key takeaways with a brief discussion and move forward to the choice-level scenario activity.
- If responses indicate confusion about why AI responded differently to text vs. image inputs, ask students to revisit AI's descriptions and reflect on what details were added or changed.
- If students struggle to connect this to effective prompting, review the warm up discussion about media choices and ask: "How does this activity confirm what we said about text vs. image information?"

🎤 Remarks

Now that we've discussed your findings, let's practice using multimodal AI in a scenario!

🖥️ Transition: Direct students to Lesson 2, Levels 5 to pick one of the scenarios to practice prompting the AI for suggestions using images and text.

👥 5

Practice with a Multimodal Model

🔄 Circulate: Ensure students recognize how AI's response shifts when using text vs. images. If students only describe objects AI listed, ask: "Did AI provide any deeper insights beyond what you expected?"

💡 Teaching Tip ▲

Encourage students to experiment with different combinations of text, images, and PDFs to see how AI's responses change. If students finish early or need an extra challenge, suggest trying Option D: Investigative Journalism, where they can analyze how AI generates headlines and summaries from different sources. Have them compare AI-generated headlines with real-world news articles on the same topic. For an added challenge, they can test how changing a single keyword, altering the image, or adding a conflicting source affects AI's response and credibility.

🗣️ Discuss: What key details did AI focus on in the image? When you prompted AI Chat with an image and a PDF, how did its response change?

💡 Discussion Goal: Students should recognize that AI was able to pull out key details from the images. Adding in a PDF helped give additional context to AI's response and make it more accurate.

💡 Teaching Tip ▲

Not all students interacted with an AI Chat level that required them to add a document. As you circulate during students' work time, find students who added both an image and a PDF to call on during the discussion.

Wrap Up (5 minutes)

☑ **Do This:** Review the lesson objectives.

☑ **Do This:** Direct students to their Unit Guides to answer the Question of the Day. Give them 30 seconds to write a response using one of the sentence starters found in the "Hint" of the Question of the Day for this lesson before having students share with a partner or ask several students to share with the class.

💡 Teaching Tip

This works well as an exit ticket or a quick way to gauge comprehension during a lesson. If time allows, guide the class discussion toward real-world applications. Use these prompts if needed:

- Where have you seen multimodal AI used outside of chatbots?
- How might multimodal AI impact careers in accessibility, medicine, or security?
- What are examples of AI tools that use images and text together? (e.g., Google Lens, AI-generated captions, AI medical imaging analysis)

✔ Assessment Opportunity

Use student responses to gauge understanding. If students struggle to explain how AI's response changed based on input type, revisit earlier examples and prompt students to compare outputs with and without images. If most responses show a strong grasp, move forward.

☑ **Do This:** Review the Key Vocabulary.



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Lesson 3: The AI's Brain

45 minutes

Overview

Career Journeys

How do AI models learn?

In this lesson, students uncover how large language models (LLMs) are trained, how they use neural networks to predict text, and how context shapes AI responses. Through an unplugged neural network activity, students simulate how AI generates predictions with and without context. They explore how LLMs rely on pattern recognition, probability, and human feedback to improve over time. Finally, students experiment with zero-shot, one-shot, and multi-shot prompting to see how different levels of input impact AI's output.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- ▶ **AP** - Algorithms & Programming

Agenda

Warm Up (10 minutes)

Activity (30 minutes)

Unplugged Neural Network

How LLMs Learn

Prompting Challenge

Wrap Up (5 minutes)

Question of the Day Comic Strip

Objectives

Students will be able to:

- Analyze how training data shapes AI behavior and biases.
- Evaluate how AI's learning affects accuracy and reliability.
- Experiment with AI responses to identify learning patterns.
- Explain how AI models, especially LLMs, use neural networks and training data.

Preparation

- Print copies of the Unplugged Neural Networks Activity Guide (one for each student)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **[KEY] Problem Solving with AI Unit Guide Exemplar** - Answer Key

▼ Make a Copy

For the students

- **How Chatbots and Large Language Models Work** - Video
- **Problem Solving with AI Unit Guide** - Activity Guide ▼ Make a Copy
- **Unplugged Neural Networks** - Activity Guide ▼ Make a Copy

Vocabulary

- **large language model (LLM)** - a general purpose form of AI that is trained on a very, very large body of knowledge
- **neural network** - an interconnected network that makes decisions using weights and hidden layers. It is used to represent the relationship between words
- **training** - the process of teaching an AI model by exposing it to large amounts of data, allowing it to learn patterns, improve accuracy, and refine its predictions over time
- **training data** - the structured or unstructured information, such as text, images, or code, used to train an AI model and shape its ability to recognize patterns and generate accurate responses

Teaching Guide

Warm Up (10 minutes)

 **Goal of this activity:** To help students understand how patterns emerge from large sets of data and how those patterns influence predictions, similar to how neural networks function.

Remarks

Imagine being in the world's biggest group chat. If you had access to every message ever sent, what might you notice? What would happen if you read them all?

 **Discuss:** *What would happen if you read every message ever sent in a giant group chat?*

 **Discussion Goal:** Students recognize that they would start noticing patterns, commonly used emojis, inside jokes, tone, and conversational flow. This sets up the idea that prediction is based on what comes before—just like how AI models work.

 **Discuss:** Click through the animated slide to display the prompts.

- *What patterns might you pick up over time?*
- *Could you start predicting what people might say next? How?*
- *How might your responses be shaped by what you've read, rather than your own original thoughts?*

 **Discussion Goal:** Students should recognize how behavior can become patterned by what they've read—just like AI does. Reinforce that:

- AI models also read (or are trained on) millions of examples.
- AI doesn't "think," it predicts based on pattern recognition.

- The more data it has, the better it becomes at mimicking human-like responses.

Remarks

Now that we've seen how AI can predict responses based on patterns in a group chat, let's take that concept further and explore how AI models, like neural networks, make predictions with limited context and how providing more context can help improve the accuracy of those predictions.

- ☒ **Do This:** Introduce the lesson objectives.
- ☒ **Do This:** Introduce the Question of the Day.

Activity (30 minutes)

Unplugged Neural Network (10 minutes)

 **Goal of this activity:** To demonstrate how neural networks predict text based on limited context and show how increasing context improves the accuracy of predictions.

☒ **Do This:** Place students in groups of three to five.

 **Distribute:** Give each group one copy of the Unplugged Neural Networks activity guide, placed face down. Let students know they'll flip it over when instructed.

☒ **Do This:** Introduce the activity. Make sure students understand that the purpose is to simulate how an AI model makes predictions with limited and full context.

Remarks

In this activity, your group will simulate how an AI model predicts words. In Round 1, you'll only get to see the last three words of a sentence before predicting the next word. That's how older models worked—with very limited memory. In Round 2, you'll get to see the entire previous sentence before making your prediction—like modern transformer models, which work with much more context. Let's see how that changes your predictions.

☒ **Do This:** Direct students to complete Round 1 in their activity guide. Use the steps on the slide to explain the activity procedure and answer any questions. When everyone has added a word, instruct student to unfold the paper and read the full sentence aloud as a group. Have groups to compare their sentences in order to see how similar or different they were. For example, students' sentences could end up like this:

- Space travel is
- exciting
- and
- dangerous
- animals
- live.

 **Circulate:** Circulate to ensure students aren't peeking back at earlier words or the full sentence.

☒ **Do This:** Direct students to complete Round 2 in their activity guide. Use the steps on the slide to explain the activity procedure and answer any questions. When everyone has contributed, read the full sentence aloud as a group. Direct groups to compare their sentences to Round 1—how did having more context change the predictions?

Example:

- The exhausted student left school to go home to . . . *rest before starting homework.*
- The exhausted student left school to go home to . . . *eat dinner with the family.*

 **Circulate:** As students read full sentences, listen for students noticing how more words made it easier to choose a logical next word. If students aren't making the connection, ask: "What helped you decide which word to add this time?" and "Was it easier or harder than the first round?"

 Teaching Tip

Due to the variety of answers possible, there is no exemplar provided for this activity. Reference the examples above for possible student generated sentences.

 **Discuss:** Click through the animated slide to display the prompts.

- *How did having limited context affect predictions?*
- *What changed when you had access to the entire starting sentence and all words that were added before you?*

 **Discussion Goal:** Students recognize that limited context leads to less accurate and more random predictions, while access to more context helps predictions become more logical and connected to previous content. This mirrors how modern AI, like transformers, improves its output by analyzing longer sequences of data.

 **Discuss:** *In the neural network activity, predictions improved when more context was provided. How do you think this applies to AI prompts?*

 **Discussion Goal:** Students recognize that just like their own predictions, AI performs better when it has more context. The more details and structure we give in a prompt, the more accurate, relevant, and helpful the AI's response becomes. Encourage students to connect this back to their experience seeing the full sentence in Round 2.

 **Do This:** Have students look at this slide. Explain that generative AI uses probabilities to predict what word should come next in a sentence. In this example, AI is most likely to predict the word "presents" or "eyes" next. AI is extremely unlikely to choose the word "clouds", which is why the percentage is so small. Since this model has limited context, i.e. it only sees the words "opened their", sometimes its predictions don't make sense in the context of what came before it.

 Teaching Tip

Make sure students understand that the percentages represent how likely AI is to choose that word to come next. In this example, AI is most likely to choose the word "presents" (40%) next. It is slightly less likely to choose "eyes" (30%) and much less likely to choose "mail" (18%), "door" (11%), or any other word.

 **Do This:** Have students look at the Improved AI Predictions slide and note the differences between this slide and the previous slide.

 **Discuss:** Why did AI's predictions change? How does this relate to the neuron activity?

 **Discussion Goal:** Students should be able to explain that increased context allows AI to make more appropriate predictions. Here, the inclusion of the words "The students" led the AI to make next word predictions related to a school setting.

 **Do This:** Have students take out their Problem Solving with AI Unit Guide.

 **Do This:** Define *neural network*.

How LLMs Learn (10 minutes)

 **Goal of this activity:** To help students understand how large language models (LLMs) like ChatGPT learn from data, recognize patterns, and refine responses through human feedback.

 **Video:** Show the "Chatbots and Large Language Models" video (0:00-1:20).

Questions to consider with the video:

- What is the difference between LLMs and typical neural networks?

Discussion Goal: Students recognize that typical neural networks perform specific tasks, while LLMs process vast text data to generate responses. LLMs predict words using probabilities, making them more flexible and capable of producing human-like text.

☑ **Do This:** Define *large language model*.

🎥 **Video:** Show the "Chatbots and Large Language Models" video (1:20-1:51).

Questions to consider with the video:

- Why is it important for the AI to look at longer sequences such as sentences or paragraphs?
- How does this relate to the context we provide in our prompts?

Discussion Goal: Students recognize that longer sequences provide more context, leading to more accurate predictions. They connect this to AI prompting, understanding that detailed prompts help AI generate more relevant and coherent responses.

🎥 **Video:** Show the "Chatbots and Large Language Models" video (1:51-2:30).

Questions to consider with the video:

- How does human feedback improve AI responses?
- Why is it still important to be cautious when using AI, even with fine-tuning?

Discussion Goal: Students recognize that human feedback helps AI refine responses by correcting errors and reducing biases. They also understand that, despite fine-tuning, AI can still produce inaccurate or biased outputs, requiring users to think critically about its responses.

☑ **Do This:** Define *training data*.

☑ **Do This:** Define *training*.

Prompting Challenge (10 minutes)

🎯 **Goal of this activity:** To give students hands-on practice with AI prompting, focusing on how context, examples, and structured inputs improve AI-generated responses. Students will explore real-world career scenarios and experiment with zero-shot, one-shot, and multi-shot prompting to refine AI responses.

🎤 Remarks

You've seen how AI predictions get better when given more context. Now, you're going to test that idea by giving the AI different types of prompts—starting with no example, then adding one, and finally, adding multiple examples. Try to notice how the responses change at each level.

🖥️ **Transition:** Direct students to Lesson 3, Level 1 on Code Studio to complete Levels 1 through 3 of the Prompting Challenges activity. For each level, students will choose one of three options based on a real-world career path:

- **Level 1:** Students provide a basic, unstructured prompt for their chosen career path.
- **Level 2:** Students add one example to guide the AI's response.
- **Level 3:** Students refine the prompt with multiple details and structure to improve accuracy.

👥 1-4

Prompting Challenges

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Circulate: Ensure students are completing the Prompting Challenges section of their Unit Guide after each level, noting how the AI's responses changed and what role context played in improving them. Listen for how they describe the differences between each AI response as they add more context. Encourage students to be specific in their reflections. If students are unsure what to write, ask:

- What changed in the AI's response when you gave an example?
- Did the response get more accurate, creative, or specific? How do you know?
- What kind of details made the biggest difference?

Teaching Tip

Each level includes multiple career path options. If time allows, encourage students to try more than one scenario to see how prompting techniques apply across different contexts. Ask them whether their observations about AI responses stay consistent across tasks or change depending on the topic.

Assessment Opportunity

Use student reflections in the Unit Guide to assess how well students understand the role of context and structure in AI responses. Look at their answers to the reflection questions after each level (zero-shot, one-shot, and multi-shot). Based on their responses, determine next steps:

- **If students give vague answers or say AI's response didn't change much across levels:**
 - Reteach the importance of context by reviewing the neural network activity analogy. Remind students that AI improves with more context and clearer instructions.
 - Prompt with a discussion question: "Why would AI struggle to give useful results if your prompt is too short or too open-ended?"
 - Small group discussion: Have students compare responses at different levels and identify specific changes in tone, structure, or accuracy.
- **If students show misunderstanding (e.g., they say multi-shot was worse or don't reference how examples helped):**
 - Review sample prompt chains together as a class. Ask: "What kind of details helped this version get better?"
 - Have students rewrite one of their weaker prompts from earlier levels using stronger structure or clarity and re-try it with the AI.
- **If students clearly articulate how adding structure and examples improved the AI's response:**
 - Move on without reteaching. These students demonstrate understanding of prompt engineering basics.
 - Optional Challenge: Encourage them to try a second career scenario and see if their insights apply across contexts.

Wrap Up (5 minutes)

Question of the Day Comic Strip

☒ **Do This:** Review the lesson objectives.

☒ **Do This:** Review the Question of the Day and give students a moment to just think about what they learned today. Let students know that they will be answering the question a little differently today.

☒ **Do This:** Direct students to create a short comic strip in their Unit Guide that illustrates something they learned about how AI learns, predicts, or responds to prompts, encouraging humor or creativity. Let students know that this is a creative way to synthesize the lesson. They can be funny, serious, or weird—as long as it's related to how AI works.

 **Circulate:** Look for students who incorporate ideas like limited context, prediction, pattern recognition, or prompting. If students are stuck, suggest they focus on one concept (e.g., how AI uses training data, how it predicts the next word, or how adding more context makes its output better).

☒ **Do This:** Review the Key Vocabulary.

☒ **Do This:** Review the Key Vocabulary.



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Lesson 4: Smart or Just Predictable?

45 minutes

Overview

Career Journeys

Where does AI excel vs struggle?

Students explore the strengths and weaknesses of AI by comparing AI-generated responses with human responses for different tasks. They learn what AI hallucinations are and how to be critical of AI-generated output. Through hands-on activities, discussions, and real-world applications, students assess AI's reliability and determine when human oversight is necessary. By the end of the lesson, students justify when AI should or shouldn't be used in various professional contexts, developing a critical mindset toward AI's role as a collaborator.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- IC - Impacts of Computing

Agenda

Warm Up (10 minutes)

AI Reliability Voting

Activity (30 minutes)

AI Hallucination Example and Discussion

AI vs Human Tasks

Wrap Up (5 minutes)

Question of the Day Reflection

Objectives

Students will be able to:

- Compare AI and human performance to identify AI's strengths and weaknesses.
- Evaluate AI's reliability and when human oversight is needed.
- Justify when AI should or shouldn't be used in real-world cases.

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **[KEY] Problem Solving with AI Unit Guide Exemplar** - Answer Key

▼ Make a Copy

For the students

- **Problem Solving with AI Unit Guide** - Activity Guide

▼ Make a Copy

Vocabulary

- **hallucination** - when the model makes up false or misleading information that sounds real but isn't true

Teaching Guide

Warm Up (10 minutes)

AI Reliability Voting

 **Goal of this activity:** Get students thinking about AI's reliability and how much human involvement they would still want in a task that can now be done by AI.

 **Say:** AI is being integrated into many real-world tasks. Some people are excited about this, while others have concerns.

 **Do This:** Inform students that they will be presented with different scenarios on the next few slides. They should think critically about the level of AI involvement they would prefer or be comfortable with for each scenario, then vote with their fingers.

Teaching Tip

If time allows, you can instead have students line up in your classroom from "Just Human" on one end and "Just AI" on the other and give them a minute to discuss their reasons for where they lined up along the spectrum with those around them.

 **Do This:** Present the first real-world AI use case: diagnosing medical issues. Ask students to vote on the level of AI involvement they would prefer for this use case. After students vote, ask a couple students from each option (0-1 finger, 2-3 fingers, 4-5 fingers) to explain their reasoning. Students should recognize that AI has strengths but also clear limitations that require human oversight. The goal is to try to get students to begin considering what makes AI reliable vs. risky in different applications.

Teaching Tip

If students struggle to think critically about the risks vs. benefits, prompt them with follow-up questions, such as "What's the worst-case scenario if AI makes a mistake in this task?" or "How could AI and humans work together in this situation?"

 **Do This:** Present the second real-world AI use case: grading essays. Ask students to vote on the level of AI involvement they would prefer for this use case. After students vote, ask a couple students from each option to explain their reasoning.

Teaching Tip

If students say AI isn't useful for grading, ask: "Could AI help teachers save time?" or "If AI misses deeper meaning in writing, how could we still use it effectively?" Bring up tools like Grammarly—AI helps with mechanics, but teachers provide feedback on ideas.

 **Do This:** Present the third real-world AI use case: driving a school bus. Ask students to vote on the level of AI involvement they would prefer for this use case. After students vote, ask a couple students from each option to explain their reasoning.

Teaching Tip

If students say AI should fully replace bus drivers, ask: "What happens if a child runs into the street unexpectedly?" or "Can AI handle a traffic situation as well as a human driver?" Connect this to real-world examples like Tesla's autopilot mode—it assists but doesn't replace drivers.

☑ **Do This:** Introduce the lesson objectives.

☑ **Do This:** Introduce the Question of the Day.

Activity (30 minutes)

AI Hallucination Example and Discussion (10 minutes)

🎯 **Goal of this activity:** Introduce students to AI-generated hallucinations and why AI gets things wrong.

☑ **Do This:** Ask students to analyze where AI went wrong. Prompt them to identify errors in AI's response and discuss why they happened. Encourage students to look for patterns in the errors. Give students a minute to share with a partner before sharing as a class.

🔄 **Circulate:** Listen for students pointing out how AI generates words based on patterns rather than understanding. If students struggle to identify why "Micken" or "Beta" is incorrect, ask: "What pattern do you see in AI's mistakes? How does AI choose words?"

💬 **Discuss:** Click through the animated slide to display the prompts.

- *How do you know AI's response is incorrect?*
- *What does this tell us about how AI generates responses?*
- *How does this show the importance of fact-checking AI outputs?*

💡 **Discussion Goal:** Students should understand that AI predicts based on probability, not actual knowledge. The goal is to get students to begin considering why AI makes errors and when it should be fact-checked. Example student responses may include:

- "AI doesn't actually know meanings; it just predicts words that commonly follow each other"
- "It made up words because it's trying to complete a pattern rather than understanding the meaning"
- "That means AI can make errors in more serious situations too, like if it made up medical information."

💡 Teaching Tip

If students are struggling to articulate why AI makes mistakes, prompt them with: "Does AI understand language like a human, or is it just following patterns?" or "What would happen if we trusted AI without checking its facts?"

☑ **Do This:** Have students take out their Problem Solving with AI Unit Guide.

🎤 Remarks

AI doesn't "think" like a human – it predicts based on probability and past data. AI can make up words, facts, or quotes because it's trying to complete a pattern, not verify truth. This occurs when the most probable response isn't accurate or factual.

☑ **Do This:** Define *hallucination*.

💡 Teaching Tip

Help students connect this definition to real-world AI failures. Explain that AI doesn't know "Micken" isn't a real word—it just predicts what's most likely to come next based on training data. This is why AI sometimes generates confident but completely false responses.

Do This: Click through the animated slide to explain that AI doesn't verify information before providing a response - it just outputs what it predicts is most likely to sound correct. This is why humans should fact-check any AI-generated responses before trusting them, even when the hallucination seems plausible. Not fact-checking can lead to misinformation.

Teaching Tip

You do not need to go in-depth into misinformation and fact-checking as this will be covered in an upcoming lesson. Use this as an opportunity to get students to start thinking about the importance of fact-checking.

AI vs Human Tasks (20 minutes)

Goal of this activity: Allow students to experiment with AI capabilities and determine where AI performs well vs. where it struggles. Encourage students to compare AI-generated vs. human-generated responses to develop the Critical Consumer mindset.

Remarks

We just saw an example of AI making up words. Now, we're going to test AI with different types of tasks to see what it does well and where it struggles.

Do This: Group students into pairs for the purpose of having a thought partner to share experiences and ranking with for the next activity.

Transition: Direct students to Lesson 4, Level 1 to complete Levels 1 through 3 to test AI's performance on different tasks. Students should complete the "Human vs AI Response Comparison" section of their Unit Guide as they interact with each task. Have students also complete the quick question on Level 4.

- **Level #1:** Students test AI's ability to solve a math problem. They rate how well a human would do and how the AI did.
- **Level #2:** Students test AI's ability to answer a creative or tricky question. They rate how well a human would do and how the AI did.
- **Level #3:** Students test AI's ability to provide historical or technical information. They rate how well a human would do and how the AI did.
- **Level #4:** Students answer a quick questions related to careers in computing science.



1-4

AI vs Human Tasks

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Circulate: Ensure students are sharing and comparing results and rankings for each task with their thought partner. Observe how students are ranking AI vs human performance in the AI vs Human Response Comparison. If students are unsure, ask: "What criteria are you using to decide whether AI or a human is better at this task?"

Teaching Tip

Each level has multiple options for students to choose from as tasks to provide to the AI. If time allows, encourage students to test multiple types of tasks to see if their rankings remain consistent.

 **Discuss:** Click through the animated slide to display the prompts.

- *What tasks was AI best at? Why?*
- *Where did it struggle? Why?*

 **Discussion Goal:** Students should recognize that AI excels at structured, data-driven tasks but struggles with creativity and reasoning. Aim to get students to connect AI's limitations to its training data and probability-based responses.

 Teaching Tip

If students struggle to identify patterns in AI's performance, ask: "What did AI do well across multiple tasks?" or "Did AI struggle with logic-based tasks or tasks that required creativity?"

☒ **Do This:** Explain that AI is really good at things like math, but it doesn't have human intuition or logic. Due to hallucinations, it needs to be fact-checked.

 Teaching Tip

If students assume AI is better just because it's faster, ask: "When does accuracy matter more than speed? Can AI make fast but incorrect decisions?"

 **Discuss:** Click through the animated slide to display the prompts.

- *How did AI's responses compare to human-generated responses?*
- *Would you trust AI's responses without fact-checking?*
- *Why or why not?*

 **Discussion Goal:** Students develop the Critical Consumer mindset by questioning where AI-generated content needs human oversight.

 Teaching Tip

If students appear to struggle to make the connection between hallucinations and the tasks they tested, ask the follow up questions: "How does what we just tested compare to the hallucination example? Did AI make similar mistakes?"

Remarks

Now that we've ranked AI's performance, let's think about how this applies to real-world careers. Do we trust AI to handle critical tasks in medicine, law, travel, or writing?

 **Transition:** Direct students to Lesson 4, Level 5 to complete the formative assessment.

☒ **Do This:** Display students' anonymous responses and ask students "What do you notice?" and "What do you wonder?" Have students discuss with a partner before asking for a couple volunteers to share.

💡 Teaching Tip

If students only focus on AI's benefits, ask: "What are the risks if AI makes a mistake in this career?" or "Would an AI error have minor or serious consequences in this field? Why?"

✔ Assessment Opportunity

This level can be used as a formative assessment to check student understanding of the lesson objectives. Here are things to look for and actions you can take:

- If students assume AI can fully replace humans, revisit the AI vs. Humans – Strengths and Weaknesses slide to reinforce AI's limitations.
- If students struggle to justify their reasoning, prompt them to connect their responses back to their AI vs. Human Response Comparison rankings.
- If students demonstrate strong understanding, move on to the wrap-up discussion.

Wrap Up (5 minutes)

Question of the Day Reflection

☑ **Do This:** Review the lesson objectives.

🎯 **Goal of this activity:** Ensure all students reflect on AI's strengths and weaknesses as a collaborator by sharing their insights in small groups. Reinforce the Critical Consumer mindset by encouraging students to verbalize when AI is useful vs. when human reasoning is needed.

☑ **Do This:** Direct students to answer the Question of the Day in their Unit Guide.

✔ Assessment Opportunity

If students struggle to differentiate AI's strengths from its weaknesses, revisit the AI vs. Humans – Strengths & Weaknesses slide and discuss specific examples. If students make strong, specific comparisons, move on to the Pass the Mic activity.

☑ **Do This:** Have students form small groups of three or four.

☑ **Do This:** Explain that now that they have reflected in their Unit Guide, they will share key takeaways with their group. Explain that the "Pass the Mic" discussion is designed so that everyone gets a chance to share. Each person will share their answer to the Question of the Day in one or two sentences and then "pass the mic" so that everyone has a turn.

🔄 **Circulate:** As students share in their groups:

- Listen for common themes and misconceptions such as:
 - Are students clearly identifying AI's strengths and weaknesses?
 - Are students misunderstanding AI's capabilities (e.g., assuming AI "thinks" like a human)?
- Prompt deeper thinking with guiding questions such as:
 - Did AI ever surprise you with what it could or couldn't do?
 - Would you trust AI more or less after today's lesson?
 - Did anyone in your group have a different perspective that made you think differently?

💡 Teaching Tip

If time allows, facilitate a full-class discussion of the QotD. Consider asking students to raise their hands if they heard a new perspective in their group that made them think differently.

- If students only state a general opinion (e.g., "AI is good at math"), follow up with: "What makes AI good at that task? Is it better than a human?" or "Can you give an example of when AI would fail at that same task?"
- If some students are not engaging in discussion, ask them directly: "What did you write down in your Unit Guide? Can you share one thing that stood out to you?" or "Did you hear something from your group that you agree or disagree with?"

☒ **Do This:** Review the Key Vocabulary.



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Lesson 5: Uncovering Contradictions

45 minutes

Overview

Career Journeys

Can AI contradict itself?

Students will be asked to explore the reasons why AI might provide contradictory answers. They will practice refining prompts in order to both induce and eliminate contradictions in AI-generated responses. Additionally, they will examine ways in which AI contradictions can have real-world effects.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- CS - Computing Systems

Agenda

Warm Up (10 minutes)

Two Truths and a Lie

Activity (30 minutes)

Contradictions in Real-World Contexts

AI Contradictions in Practice

Wrap Up (5 minutes)

Future Predictions

Objectives

Students will be able to:

- Analyze how AI's reliance on probability-based models can lead to inconsistencies.
- Evaluate the implications of AI-generated contradictions in real-world applications.
- Experiment with AI to uncover contradictions and explain why they occur.

Preparation

- Print copies of the Examine AI Contradictions activity guide (one for each pair of students)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **[KEY] Problem Solving with AI Unit Guide Exemplar** - Answer Key

▼ Make a Copy

For the students

- **Examining AI Contradictions** - Activity Guide ▼ Make a Copy
- **Problem Solving with AI Unit Guide** - Activity Guide ▼ Make a Copy

Vocabulary

- **context** - additional information or data surrounding a prompt or query

that helps the AI generate more accurate and relevant responses

- **contradiction** - when an AI generates conflicting or opposing responses based on a similar input, making it impossible for both responses to be correct at the same time

Teaching Guide

Warm Up (10 minutes)

Two Truths and a Lie

☑ **Do This:** Ask the class for a topic (e.g., space, animals, history, music, etc.).

☑ **Do This:** Use Lesson 5, Level 1 to demo entering the following prompt: "Tell me 2 truths and a lie about [insert topic]. Don't tell me which ones are truths or lies."



Two Truths and a Lie

☑ **Do This:** Have students partner up and try to guess which AI-generated statements are true and which is the lie.

Remarks

AI might not always produce the results that is expected. The result might even be contradictory. Therefore, fact-checking AI's results is an important step of the process.

☑ **Do This:** Demonstrate fact-checking by researching online using Google and credible websites to verify which statements are true and which is a lie.

 **Discuss:** Click through the animated slide to display the prompts.

- *Why might AI generate a lie or inaccurate information?*
- *How do we ensure accuracy when using AI?*

💡 **Discussion Goal:** Students should understand that AI generates responses based on patterns in data rather than verified facts. They should discuss how AI lacks direct knowledge, can misinterpret ambiguous prompts, and sometimes "hallucinates" information because it is designed to predict plausible-sounding text rather than confirm truth. Students highlight the importance of human judgement in verifying AI-generated content. Students should identify fact-checking strategies, such as cross-referencing multiple reliable sources (using official or expert-verified information) and critically analyze responses for logical consistency.

☑ **Do This:** Introduce the lesson objectives.

☑ **Do This:** Introduce the Question of the Day.

Activity (30 minutes)

Contradictions in Real-World Contexts (15 minutes)

☑ **Do This:** Define *contradiction*.

📄 **Distribute:** Give each pair of students a copy of the Examining AI Contradictions activity guide.

📄 2

Examining AI Contradictions Activity Guide

☑ **Do This:** Using their activity guide, students will review two contradictory AI responses from a telehealth chatbot.

- **Step 1:** Students review both AI-generated responses.
- **Step 2:** Students highlight contradictions, vague statements, or misleading claims, such as the conflicting advice on whether or not immediate medical attention is necessary.
- **Step 3:** Students fact-check the AI's claims using reliable sources (e.g., trusted medical websites or professional guidelines).
- **Step 4:** Students refine the provided prompt to get a more consistent and clearer response from the AI.

💬 **Discuss:** Click through the animated slide to display the prompts.

- *How does the AI's lack of specialization in the medical field contribute to contradictions?*
- *How can fact-checking and prompt refinement help improve AI-generated responses in real-world contexts?*

💡 **Discussion Goal:** Students should recognize that AI models do not have deep expertise like medical professionals. They should discuss how AI relies on broad datasets that may contain conflicting or outdated information, which leads to contradictions. Students should understand that fact-checking ensures accuracy by verifying AI-generated information against trusted sources. They should also recognize that refining prompts (e.g. specifying a reliable source, adding context, etc.) helps AI produce clearer and more relevant responses.

🎤 Remarks

AI models do not have deep expertise like medical professionals. AI relies on broad datasets that may contain conflicting or outdated information, which leads to contradictions. Fact-checking ensures accuracy by verifying AI-generated information against trusted sources. Refining prompts by doing things like specifying a reliable source or adding context helps AI produce clearer and more relevant responses.

☑ **Do This:** Explain the differences between where an AI and a medical professional acquire information.

AI Contradictions in Practice (10 minutes)

🖥️ **Transition:** Direct students to Lesson 5, Level 3 to complete Levels 3 and 4. Students choose a different topic in this choice level (e.g., government, technology, education, social media). In pairs, students practice with the various forms of questioning provided and discuss the results. Have them add their reflections to the last page of their Activity Guide.

👥 3-4

AI Contradictions in Practice

3

4

☑ **Do This:** Click through the animated slide to review the three different ways in which students were asked to refine the prompt in the previous activity.

☑ **Do This:** Define *context*.

🖥 **Transition:** Direct students to Lesson 5, Level 5. In pairs, students practice prompting the AI Chat with different framings of their own questions to see if they can produce AI contradictions and can experience themselves how their framing can alter the response an AI outputs.

💡 Teaching Tip

If students aren't sure what topic to choose, direct them to choose one of the examples from the level:

- Mandatory physical education in school
- Free higher education
- Use of AI in school assignments
- The current US voting age

If students need help with prompting the AI, direct to them to modify the prompt examples from the level:

- Positive framing
- Negative framing
- Positive framing with a reliable source



5-6

Try Your Own

5

6

☑ **Do This:** Direct students to Lesson 5, Level 7 to complete the formative assessment on Levels 7 and 8.



7-8

Check Your Understanding

7

8

✔ Assessment Opportunity

This level can be used as a formative assessment to check student understanding of the lesson objectives. Here are things to look for and actions you can take:

- Review responses to see whether students can explain why AI might contradict itself and how that differs from human contradictions. If students struggle to articulate these differences, have them revisit the telehealth example (or another real-world scenario) and prompt them to highlight exactly where the AI's contradiction arose—and contrast it with how a human might revise a diagnosis after new data.
- See whether students can identify specific potential harms, and if not, direct them back to the telehealth chatbot examples provided in the lesson. You could also show short clips or articles about AI's role in other high-stakes fields (finance, insurance) to further underscore the importance of verifying AI outputs.

Wrap Up (5 minutes)

Future Predictions

- ☒ **Do This:** Review the lesson objectives.
- ☒ **Do This:** Review the Question of the Day.
- ☒ **Do This:** Have students take out their Problem Solving with AI Unit Guide.
- ☒ **Do This:** In their Unit Guide, students answer the Question of the Day.
- ☒ **Do This:** Students make a prediction about how the ability of AI to contradict itself might evolve over the next 10 years.
- ☒ **Do This:** Review the Key Vocabulary.



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Lesson 6: Understanding Bias

45 minutes

Overview



What is AI bias, and how does it emerge?

In this lesson, students explore how AI bias emerges from biased training data. Students analyze results from online search engines and AI Chat to examine them for bias. In the main activity, students select candidates for a particular job by filtering resumes. They compare their selections with those made by AI and note any bias in the selection process. Throughout the lesson, students brainstorm and discuss strategies to make AI systems more fair and ethical.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

► IC - Impacts of Computing

Agenda

Warm Up (10 minutes)

AI Assumptions

Activity (25 minutes)

Bias Intro: Career Clothing

AI Audit Challenges

Wrap Up (10 minutes)

3-2-1 Reflection and Discussion

Objectives

Students will be able to:

- Analyze AI-generated responses to identify bias and assess its impact on fairness.
- Experiment with AI prompts and data inputs to explore how they affect AI bias.
- Explain how AI bias emerges from training data and influences AI-generated outputs.
- Identify strategies for mitigating bias in AI systems and evaluating ethical considerations.

Preparation

- Print copies of the AI Audit activity guide (one for each student)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **[KEY] Problem Solving with AI Unit Guide Exemplar** - Answer Key

▼ Make a Copy

For the students

- **AI Audit** - Activity Guide
- **Problem Solving with AI Unit Guide** - Activity Guide

▼ Make a Copy

▼ Make a Copy

Vocabulary

- **AI bias** - when a model favors certain information over others
- **alignment** - ensuring an AI system's goals and outputs match human values, fairness, and safety

Teaching Guide

Warm Up (10 minutes)

AI Assumptions

Remarks

AI models learn by analyzing vast amounts of data. They recognize patterns and use them to make predictions. This is an example of an AI that is trained to play rock, paper, scissors. It has been trained on images of hands and has to predict what the user played.

☒ **Do This:** Ask students if AI would label this image as rock, paper, or scissors? Discuss how it looks similar to other hands in the "Paper" category so AI would likely label this as "Paper" as well.

Teaching Tip

Ask students to identify patterns in the data. For example, the AI Bot might recognize this hand as "Paper" because it sees 5 outstretched fingers. It might recognize that "Scissors" has two outstretched fingers, and "Rock" has none.

 **Discuss:** Ask students how AI might label this image.

 **Discussion Goal:** Explain that AI might incorrectly label this image because it was not trained on images of hands held up in the air and was only trained on images of hands straight out. This is an example of bias in the training data, because the AI Bot was only trained to recognize hands from certain angles.

 **Discuss:** What, or who, is not represented by these images? Give examples of some images that should be added to this dataset to improve the AI model.

 **Discussion Goal:** Students should understand that AI models learn patterns and replicate biases. In order to to reduce or eliminate bias, more images should be added to the training dataset. Examples of images that should be added include, hands from different angles, hands missing fingers, hands with rings or other jewelry, etc.

Teaching Tip

If you have time, test out this trained model from Teachable Machine:
<https://teachablemachine.withgoogle.com/models/fadF6RxOcF/>.

This model was built with Kaggle data from thousands of pictures of hands:
<https://www.kaggle.com/drgfreeman/rockpaperscissors>. Note, this dataset primarily features images of the back of hands held straight up in the air and shows bias against hands viewed from the side or from the palm.

As an extension activity, consider creating your own Rock, Paper, Scissors model using Teachable Machine:
<https://teachablemachine.withgoogle.com/>

- ☑ **Do This:** Introduce the lesson objectives.
- ☑ **Do This:** Introduce the Question of the Day.

Activity (25 minutes)

Bias Intro: Career Clothing (10 minutes)

🎯 **Goal of this activity:** Students explore AI's assumptions and analyze how training data influences AI-generated responses. Students analyze AI's initial responses, identify potential biases, and refine their prompts to create more neutral and inclusive descriptions.

- ☑ **Do This:** Have students take out their Problem Solving with AI Unit Guide.

🎤 *Remarks*

AI learns from patterns in data, but that data isn't always balanced or fair. If certain perspectives or groups are overrepresented, AI will reflect those patterns in its responses. This can lead to AI making assumptions that don't accurately represent the full picture.

- ☑ **Do This:** Define *AI bias*.

💡 Teaching Tip

If you choose to demonstrate doing a Google Image search for "nurse" or "doctor", consider searching for animations, e.g. "nurse cartoon image", to see more obvious bias.

- ☑ **Do This:** Place students in pairs.

📄 **Distribute:** Give each pair of students a copy of the Understanding Bias activity guide.

🎤 *Remarks*

AI generates responses based on patterns in its training data. But what happens when those patterns reflect assumptions about people and professions? Let's investigate by asking AI to describe outfits for two different careers and analyzing its responses.

💻 **Transition:** Direct students to Lesson 6, Level 1 to explore AI-generated descriptions of clothing worn by different professions – a secretary and a CEO. Students should complete the AI Bias Analysis section of their activity guide as they interact with each task.

💻 1

Career Clothing

💡 Teaching Tip

It might be hard to see overt bias in AI Chat's responses. Consider discussing any nuanced biases which exist in society that are being reflected by the chatbot. For example, the chatbot returns style suggestions for both male and female CEOs but the suggestions differ by gender. Female CEOs are encouraged to wear skirts, dresses, heels, and flats but not chinos, button down shirts, or loafers. This reflects bias in our societal norms

that the chatbot is parroting back to us.

 **Discuss:** Click through the animated slide to display the prompts.

- *Did AI assume a default identity in either of the prompts?*
- *Did the responses differ in the tone, assumptions, or characteristics included?*

 **Discussion Goal:** Students should recognize that AI descriptions reflect patterns in training data, which may reinforce stereotypes or assumptions. They should identify differences in language, gender associations, or implied expertise in AI-generated responses.

Teaching Tip

If students struggle to articulate differences, prompt them with:

- "What words or phrases stand out in each description?"
- "Does AI use the same level of professionalism, detail, or authority in both responses?"
- "Would changing the wording of the prompt lead to a different response?"

Remarks

Now that we've seen how AI describes outfits for different professions, let's take it a step further. AI doesn't generate biased responses on purpose—it reflects patterns in the data it has learned from. But we can adjust our inputs to get better, more neutral results. Let's try to refine our prompts and see how AI responds.

☒ **Do This:** Return students to Lesson 6, Level 1. Refine the original prompts to improve neutrality. Students document their refined prompts and compare AI's new responses in the AI Bias Analysis Chart section of their activity guide.

 **Discuss:** *What factors in AI's training data may have led to this bias we uncovered?*

 **Discussion Goal:** Listen for students to bring up bias in the datasets, connecting back to the warm up activity, other examples they may be familiar with, or their own lived experiences.

Remarks

We've explored how AI bias emerges and how we can adjust prompts to reduce it. Now, let's think about what AI developers can do to make AI systems more fair and representative.

 **Transition:** Direct students to Lesson 6, Level 2 to complete the formative assessment.

☒ **Do This:**

- Students answer the multiple-select question: "How could AI developers make adjustments to reduce bias?"
- Use the "View Student Responses" feature to analyze the class results.
- Ask students: "What do you notice?" and "What do you wonder?" Have them discuss with a partner before selecting a few volunteers to share.

 2

 Check Your Understanding

Teaching Tip

Emphasize that AI bias is not just a coding issue—it's a data issue. Many students assume developers can "fix"

bias with a simple programming change, but bias originates from the datasets AI is trained on.

✓ Assessment Opportunity ▲

This activity helps assess student understanding of AI bias mitigation strategies. Based on student responses, take the following actions:

- If most students select the correct answer (B), proceed to the next activity (AI Audit).
- If some students are confused, have them pair up and explain their reasoning to each other. Then, re-pose the question and have students show their answers using their fingers (1-4) before moving on.
- If many students select incorrect answers (A, C, or D),
- Revisit key slides that explain how bias comes from data, not just code.
- Clarify that more data isn't always better—biased data will still lead to biased AI.
- Have students answer the question again using a show of fingers before transitioning.

AI Audit Challenges (15 minutes)

 **Goal of this activity:** Students analyze bias in decision-making by comparing their own candidate selection process with AI's and exploring how training data influences AI's choices.

Remarks

AI is increasingly being used to make important decisions—hiring employees, admitting students, and even determining which patients receive medical care first. But how do we know if AI is making fair and unbiased choices? Today, you'll take on a decision-making role and compare your choices to AI's.

☒ **Do This:** Direct students to Lesson 6, Level 3. Students choose one decision-making role:

- **Hiring Manager:** Selects top candidates for job interviews.
- **College Admissions Officer:** Picks students for acceptance.
- **Hospital Triage Nurse:** Prioritizes patients for urgent care.

Then students review the candidate pool and select two candidates based on their qualifications. They record and justify selections in the activity guide.



Decision-Making Roles

 **Circulate:** Listen for students discussing why certain candidates stood out over others. If students struggle, ask:

- *"What specific qualifications made you choose these candidates?"*
- *"Did you consider factors beyond qualifications, like personal background?"*

Remarks

Now, let's see how AI makes the same decision. AI doesn't have opinions or instincts—it relies on patterns in data. You'll ask AI to select candidates for your decision-making role, analyze its reasoning, and compare it to your own. Finally, you'll refine your prompt to see if AI's choices can better match your priorities.

☒ **Do This:** Direct students to Lesson 6, Level 4, where they will ask AI Chat to select candidates for the same decision-making role they previously chose. Students will then compare AI's selections to their own. After reviewing AI's choices, students will refine their prompt to better align with their group's priorities, specifying which qualities AI should prioritize while abstracting away irrelevant details.



4-5

AI Audit Challenge

4

5

☒ **Do This:** Form student groups of three, ensuring each student has a different role, to share their findings.

☒ **Do This:** Define *alignment* and share two mitigation strategies for reducing AI bias.

Wrap Up (10 minutes)

3-2-1 Reflection and Discussion

☒ **Do This:** Review the lesson objectives.

☒ **Do This:** Review the Question of the Day.

☒ **Do This:** In the Unit Guide, students reflect on 3 examples of AI bias from this lesson, 2 things to help reduce AI bias, and 1 cause of AI bias. After, they discuss their responses with the class.

☒ **Do This:** Assign students numbers in small groups and give them a discussion prompt. After a short discussion, randomly call out a number—students with that number share their group's response.

💡 Teaching Tip

Rotate which number is called in future discussions to ensure all students participate over time.

☒ **Do This:** Review the Key Vocabulary.



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Lesson 7: AI's Wild Imagination

45 minutes

Overview

What are some examples of AI hallucinations? What are their impacts?

This lesson explores the real-world impacts of AI hallucinations. Building on their previous understanding of AI errors, students analyze real-world case studies where AI misinformation has caused harm. Through discussion, case studies, and fact-checking strategies, students evaluate AI's reliability and consider its ethical implications in fields like law, healthcare, and journalism.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- IC - Impacts of Computing

Agenda

Warm Up (10 minutes)

Hallucination Chain

Activity (30 minutes)

Hallucination Case Studies

Reduce Hallucinations and Assessment

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Evaluate AI-generated content for accuracy and potential hallucinations.
- Explain the potential risks and consequences of AI hallucinations in real-world scenarios.
- Identify strategies to reduce AI hallucinations.
- Recognize AI hallucinations and describe how they happen.

Preparation

- Print copies of the Hallucination Case Studies activity guide (one per student)
- Print copies of the Jigsaw Articles (one copy group of four students)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **[KEY] Problem Solving with AI Unit Guide Exemplar** - Answer Key

▼ Make a Copy

For the students

- **Hallucination Case Studies** - Activity Guide ▼ Make a Copy
- **Jigsaw Articles** - Handout ▼ Make a Copy
- **Problem Solving with AI Unit Guide** - Activity Guide ▼ Make a Copy

Teaching Guide

Warm Up (10 minutes)

Hallucination Chain

 **Goal of this activity:**Reveal how AI can generate misinformation by mimicking human knowledge gaps.

 **Do This:** Place students in groups of four.

 **Distribute:** Give each group 12-15 sticky notes. Each students needs at least three.

Teaching Tip

If needed, adjust the number of sticky notes based on group size so each student has three.

 **Do This:** Have students each write three facts about the Great Wall of China on sticky notes - one fact per sticky note. Tell students that if they are unsure, they can make up something that sounds believable.

 **Circulate:** Check that students are writing individual facts and not collaborating at this stage. If students hesitate, remind them that they can create facts if they're unsure.

 **Do This:** Have groups examine their sticky notes and categorize them as similarities, contradictions, or inaccuracies.

 **Circulate:** Watch for students who categorize everything as similar or accurate—prompt them to look for contradictions.

Teaching Tip

If students struggle, prompt them with: "How do we know if something is inaccurate?" or "What makes two facts similar?"

 **Discuss:** Click through the animated slide to display the prompts.

- *What might happen if we published these "facts" on a public website?*
- *How does this activity mimic the way that AI generates responses?*

 **Discussion Goal:** Guide students to recognize just like we created statements that sounded true but weren't, AI can hallucinate and generate misinformation that seems convincing. AI models learn from human input—if that input contains misinformation, AI can amplify and spread it.

 **Do This:** Introduce the lesson objectives.

 **Do This:** Introduce the Question of the Day.

Activity (30 minutes)

Hallucination Case Studies (15 minutes)

 **Goal of this activity:**Show how AI hallucinations cause harm in real-world applications.

 **Distribute:** Give each student a copy of the Hallucination Case Studies Activity Guide and a Jigsaw Article.

☑ **Do This:** Explain that students will read about real-world AI hallucinations in industries like law, travel, healthcare, and transportation. They will read independently first, then discuss their findings with their group.

☑ **Do This:** Explain that hallucinations exist because AI doesn't have a database of "true" facts. It predicts based on patterns in data. If it is trained on a large set of data that haven't been fact-checked, it can generate biased and inaccurate output.

☑ **Do This:** Instruct students to read their assigned Jigsaw Article individually. Each student should annotate their article by highlighting key terms (e.g., AI Hallucination, accuracy, fairness, impact) and writing notes or questions in the margins. Students should also answer the reflection questions in their Hallucination Case Studies Activity Guide as they read their article.

🔄 **Circulate:** Walk around and check that students are actively annotating and engaging with the text, jotting down notes in their activity guides. If students struggle, ask:

- "What was the AI hallucination in your article?"
- "What was the real-world impact?"
- "Would a human have made the same mistake?"

If students are only highlighting without writing notes, encourage them to add margin notes or questions to deepen their analysis.

💡 Teaching Tip

If students finish early, have them jot down a real-world example where they've seen AI generate incorrect or misleading information (e.g., a chatbot making up a statistic, an AI-generated image mislabeling an object). This helps students connect their reading to prior knowledge and real-world experiences.

☑ **Do This:** Have students share their article summaries with their group members. Each student should briefly explain what their article was about, what the AI hallucination was, and its consequences. After each student shares, the group discusses the real-world impact of AI hallucinations and what could have been done differently.

🔄 **Circulate:** Listen for common themes in student discussions. If students struggle to identify AI's impact, ask:

- "Who was affected by this AI mistake?"
- "What industries could be impacted by similar hallucinations?"
- "What steps could have been taken to avoid this hallucination?"

💡 Teaching Tip

If groups struggle to connect their article's AI hallucination to broader issues, encourage them to think about similar situations in other fields (e.g., misinformation in journalism, AI-generated legal errors, biased hiring algorithms).

Reduce Hallucinations and Assessment (15 minutes)

🎯 **Goal of this activity:** Students get practice with evaluating AI responses that contain hallucinations.

🎤 Remarks

Now that we've seen the real-world impact of AI hallucinations, let's look at three strategies we can use to minimize misinformation when working with AI.

☑ **Do This:** Click through the animated slide to explain how it helps prevent AI misinformation. Use real-world examples where possible, such as AI misquoting sources or giving outdated information. Remind students that AI doesn't fact-check—it predicts likely responses based on past data.

💡 Teaching Tip

Encourage students to connect these strategies to their own experiences using AI tools or searching for information online. If students struggle to see the relevance, ask: "What happens if you search for a fact and only look at the first result without verifying it?"

 **Transition:** Direct students to Lesson 7, Level 1 to complete Levels 1 through 2.

- **Level #1:** Students review how to reduce AI hallucinations.
- **Level #2:** Students complete a formative assessment.

 1

Reduce AI Hallucinations

 2

Check Your Understanding

 **Circulate:** Observe students as they complete the assessment. Look for signs that they understand AI hallucinations and how to counter them. Note if students struggle to connect fact-checking techniques to real-world examples.

✔ Assessment Opportunity

Use student responses to assess how well they grasp the risks of AI hallucinations and the strategies to counter them. Teachers can determine next steps based on common student responses:

- If students identify hallucinations but struggle to explain their impact:
 - Have students revisit the Hallucination Case Studies slide and discuss their notes in small groups.
 - Ask: "Why does AI misinformation matter in real-world settings?"
- If students don't apply fact-checking strategies:
 - Direct them to the Reduce AI Hallucinations slide and have them match each strategy (careful prompting, skepticism, fact-checking) to a real-world example.
 - Ask: "Which strategy would help verify this AI-generated claim?"
- If students effectively explain AI hallucinations, impacts, and fact-checking strategies:
 - Move on to the wrap-up discussion.
 - Have students share how they would explain AI hallucinations to someone unfamiliar with them.

Wrap Up (5 minutes)

 **Goal of this activity:** Engage students in critical thinking about AI hallucinations.

 **Do This:** Review the lesson objectives.

 **Do This:** Have students take out their Problem Solving with AI Unit Guide.

 **Do This:** Instruct students to take a minute to write a response to the Question of the Day in their Unit Guide. Encourage them to use specific examples from the lesson, such as case studies or AI-generated errors they encountered.

✔ Assessment Opportunity

The Question of the Day can be used as a formative assessment to check whether students can recall real-world examples of AI hallucinations and their impacts. If students struggle to provide examples, review key moments from the lesson, such as the case studies and AI panel discussion.

☒ **Do This:** Explain that students will respond to statements about AI hallucinations by standing up if they agree or staying seated if they disagree. Emphasize that there are no right or wrong answers—the goal is to encourage reasoning and discussion.

☒ **Do This:** Read each statement aloud. Pause after each statement and let students vote by standing or sitting. Ask a standing student and a seated student to share their reasoning. Use the follow-up prompts:

- *What makes hallucinations easy or hard to spot?*
- *Can you think of a real-world example where an AI hallucination was helpful or harmful?*
- *What factors make AI hallucinations dangerous in some cases but not others?*

💡 Teaching Tip

If students all agree on a statement, challenge them by asking: "Can you argue the other side?" or "What might change your perspective?"



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Lesson 8: Debugging and Refining Outputs

45 minutes

Overview

Career Journeys

Can we improve AI's responses?

Students explore how to improve AI-generated responses by refining their prompts using a five-step framework: task, context, reference, format, and tone. Through hands-on exploration in AI Chat levels, they test different prompt modifications, analyze how AI responds to varying levels of detail, and apply debugging techniques to enhance accuracy. At the end of the lesson, students summarize their learning about AI prompt refinement by adding ideas to a classroom Anchor Chart.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- CS - Computing Systems

Agenda

Warm Up (5 minutes)

What is a Prompt Engineer?

Activity (35 minutes)

AI Prompting Framework

Check Your Understanding

Wrap Up (5 minutes)

Key Elements Anchor Chart

Objectives

Students will be able to:

- Apply debugging techniques to improve AI-generated responses.
- Explain how prompt refinement improves AI outputs.

Preparation

- Prepare a classroom anchor chart containing five columns - Task, Context, Format, Reference, and Tone.

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **[KEY] Problem Solving with AI Unit Guide Exemplar** - Answer Key
▼ Make a Copy

For the students

- **AI's Hottest New Job Pays Up to \$250K a Year. So I Applied. | WSJ** - Video
- **Problem Solving with AI Unit Guide** - Activity Guide
▼ Make a Copy

Vocabulary

- **prompt engineering** - the process of creating precise and effective prompts to guide generative AI models

Teaching Guide

Warm Up (5 minutes)

What is a Prompt Engineer?

 **Goal of this activity:** To introduce an interesting new career that is related to the work that students will be asked to do today— refining prompts in order to produce high-quality AI output.

 **Video:** Show the "AI's Hottest New Job Pays Up to \$250K a Year. So I Applied" video.

Questions to consider with the video:

- *What does a prompt engineer do?*
- *What skills do you need to have to be a prompt engineer?*

Discussion Goal: Students should be able to explain that a prompt engineer is someone who develops prompts to get the best output from a large language model. A prompt engineer needs to be good at writing clearly and giving good instructions. They should know how to translate between a machine and a human.

☒ **Do This:** Define *prompt engineering*.

☒ **Do This:** Introduce the lesson objectives.

☒ **Do This:** Introduce the Question of the Day.

Activity (35 minutes)

AI Prompting Framework (25 minutes)

 **Goal of this activity:** Provide students with a structured method to improve AI-generated responses using five key elements of a strong prompt. They will have a chance to test out each method as they work through levels.

☒ **Do This:** Click through the animated slide to introduce the key elements of a prompt. Explain that just like giving instructions to a friend, the more details you provide, the better the response. By specifying what we want AI to do, giving context, and even suggesting a format, we can make AI's responses much more useful.

☒ **Do This:** Introduce "TaCo FoRT" as a mnemonic: **T**ask, **C**ontext, **F**ormat, **R**eference, **T**one.

Teaching Tip

Suggest that students remember "TaCo FoRT" so they can systematically check if they've included each element in their prompts. This prompting framework can be used in future interactions with AI Chatbots in this lesson and beyond.

Behind the Curtain: Explain to students that AI doesn't understand a user's intent. Instead, it sees words in context and simply *guesses* what should come next. Strategies like TaCo FoRT work because each element in gives an AI chatbot more structure and context from which to make its guesses.

 **Distribute:** Direct students to take out their Problem Solving with AI Unit Guide.

 **Transition:** Direct students to Lesson 8, Level 1 to practice providing the core task or objective the AI needs to address. Students should complete the corresponding section of their Unit Guide.

Circulate: As you move around the room and engage students in thoughtful reflection by asking, "How did AI's response change after you added context?" or "Did AI follow your requested format? Why or why not?" or "What's still missing from AI's response?". Offer guidance if a prompt remains too vague (e.g., "Make sure you tell the AI exactly what you want it to do!").

Discuss: Click through the animated slide to display the prompts.

- *What task did you add to the original prompt?*
- *How did refining the prompt improve AI's output?*

Discussion Goal: The aim is for students to recognize that making the Task explicit often leads to more focused and relevant AI replies. They should see how even small changes in wording can clarify the objective for the AI. This discussion should serve as an opportunity for guided practice before students work independently through the next four refining output activities.

Do This: Explain the connection between *refining a prompt* and *debugging* in traditional software development.

Do This: Direct students to Lesson 8, Level 2 to complete Levels 2 through 5. Students should complete the corresponding sections of their Unit Guide.

- **Level #2:** Students practice providing necessary background information to give context for the task.
- **Level #3:** Students practice providing references, examples, or models to guide the AI.
- **Level #4:** Students practice specifying the format in which the response should be delivered.
- **Level #5:** Students practice setting the tone for the AI response.
- **Level #6:** Students answer a quick questions related to careers in computing science.



Circulate: As you move around the room and engage students in thoughtful reflection by asking, "How did AI's response change after you added context?" or "Did AI follow your requested format? Why or why not?" or "What's still missing from AI's response?"

Do This: Click through the animated slide to review the outcomes of debugging and refining output. Ensure students understand that every time they refine their prompt, they are debugging AI—just like a programmer debugs their code. The first response might be generic, but by adjusting details and testing different versions, they can generate clearer, more accurate AI outputs.

Check Your Understanding (10 minutes)

Goal of this activity: Students refine a generic prompt using the Prompting Framework to improve accuracy.

Do This: Direct students to Lesson 8, Level 7 to complete Levels 7 through 9.

- **Level #7:** This panel level serves as a review of the key elements of prompt refinement.
- **Level #8:** Students choose a generic prompt to improve. They use at least three elements (Task, Context, Reference, Format, Tone) and iterate on their prompt until they have one that consistently generates helpful AI output.
- **Level #9:** Students copy their revised prompt from Level 8 into the free response level. Then they explain why their revised prompt led to a better result than the initial generic prompt.



7-9

Check Your Understanding

7

8

9

✓ Assessment Opportunity ▲

The assessment on Level 9 can be used as a formative assessment of the lesson objectives. As students submit their final revised prompts, circulate and provide quick feedback by:

- Looking for clear, specific refinements that improved the AI's output.
- Asking probing questions like:
 - "What made the AI's response better after refining your prompt?"
 - "How did adding context change the AI's response?"
 - "Why do you think AI initially struggled with this prompt?"

Wrap Up (5 minutes)

Key Elements Anchor Chart

🎯 **Goal of this activity:** Students summarize their learning about AI prompt refinement in a concise, creative, and engaging way.

☑ **Do This:** Review the lesson objectives.

☑ **Do This:** Review the Question of the Day.

📄 **Distribute:** Give each student a sticky note.

☑ **Do This:** Instruct students to take a minute and answer the Question of the Day on a sticky note, providing a specific example of how one of the key element (Task, Context, Format, Reference, Tone) might be used to refine output.

Example Student Responses:

- **Format:** Write a poem, such as a haiku or limerick
- **Tone:** Write in a professional tone, as if you are a college admissions counselor

🔄 **Circulate:** Prompt deeper thinking with a guiding question:

- *What strategies did you use in the levels?*
- *Give an example of how you added more context or chose a different tone.*

☑ **Do This:** Have students add their sticky notes to the classroom anchor chart.

💡 Teaching Tip ▲

Use color-coded notes for different key elements to create a more structured visual representation. This classroom anchor chart will become useful for students to try new ways to refine their prompts throughout this unit and others in the CSAIF curriculum. Encourage students to add to the anchor chart whenever they have a new way of refining their prompts that leads to more interesting, accurate, and fair output.

✓ Assessment Opportunity ▲

The Question of the Day can be used as a formative assessment of understanding the relationship between prompting techniques and improved AI-generated responses. Make sure students write their name on their sticky note if you plan to use it as an assessment.



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Lesson 9: AI as a Co-Creator

45 minutes

Overview

Career Journeys

How can AI enhance creativity?

In this lesson, students take on creative roles to design and launch a product or app that helps students succeed in school. Working in teams, they collaborate with AI across four stages—research, brainstorming, content generation, and editing—to co-create campaign materials like slogans, scripts, and visuals. The activity emphasizes decomposition, prompting strategy, and human-AI collaboration, helping students develop both technical and creative problem-solving skills.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- ▶ **AP** - Algorithms & Programming

Agenda

Warm Up (10 minutes)

AI Essay Critique

Activity (30 minutes)

Introduction to AI Collaboration

Collaborate on a Product Launch

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Analyze the strengths and weaknesses of AI-generated creative work compared to human-created content.

Preparation

- Print copies of the Collaborate on a Product Launch activity guide (one for each student)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **[KEY] Problem Solving with AI Unit Guide Exemplar** - Answer Key
▼ Make a Copy

For the students

- **Collaborate on a Product Launch** - Activity Guide ▼ Make a Copy
- **Problem Solving with AI Unit Guide** - Activity Guide ▼ Make a Copy

Vocabulary

- **co-creating** - humans and AI working together, with humans providing creativity and direction, while AI assists by processing data, generating suggestions, or automating tasks

Teaching Guide

Warm Up (10 minutes)

AI Essay Critique

 **Goal of this activity:** Have students understand the importance of keeping a human in the loop and where AI might lack creativity or insight.

 **Do This:** Direct students to take out their Problem Solving with AI Unit Guide.

 **Do This:** Direct students to the "AI Essay Critique" section in their Unit Guide to read the paragraph. Explain that the paragraph was generated by an AI using the basic prompt: "Write a personal essay". Direct students to identify what the AI did well and where a human might add more value to the writing.

 **Discuss:** Click through the animated slide to display the prompts.

- *What does this introduction do well?*
- *What is it lacking?*
- *How could a human collaborating with AI improve this draft?*

 **Discussion Goal:** Students recognize that AI produces technically correct and well-structured writing. Students also note that AI writing can lack originality, emotion, and personal voice. They suggest ideas about collaborating with AI to "fine-tune" the response, or to improving their prompts to provide more context for the original response. This activity lays the groundwork for today's project. Students will later collaborate with AI themselves to generate and refine creative content, so this is a chance to start thinking critically about how AI performs.

 **Discuss:** *If AI struggles with originality in personal essays, how might that apply to creative fields like marketing, advertising, and branding?*

- **Stand and Share:** students stand and form small groups of 3-4 students from different tables or rows. After discussing the prompt, each group picks one representative to share their ideas with the class.

 **Discussion Goal:** Students make connections between AI's limitations in personal essays to similar creative challenges in other fields. They understand that creative ideas in marketing and branding will require emotional connections, cultural relevance, and distinct voices that AI will need human input on to be truly effective.

Teaching Tip

If students struggle, prompt them with: "Would you trust AI to write a Super Bowl ad? Why or why not?" or "What makes a good slogan or ad? Could AI come up with it on its own?"

Remarks

Now that we've seen how AI can help generate ideas, but also where it might fall short, let's dive into how we can collaborate with AI to create something truly innovative—where humans and AI can work together to refine and enhance creative projects.

 **Do This:** Introduce the lesson objectives.

 **Do This:** Introduce the Question of the Day.

Activity (30 minutes)

Introduction to AI Collaboration (10 minutes)

 **Goal of this activity:** In this activity, students will work in small groups to design and launch a new product or app intended to help students with school-related tasks. They will collaborate with AI at each stage of the development process, refining ideas and generating content.

 **Discuss:** Click through the animated slide to display the prompts.

- *More and more music and movie producers, graphic designers, writers and other creatives are taking advantage of collaborative opportunities with AI. How might AI help you in one of these careers?*
- *Where might AI fall short?*

 **Discussion Goal:** Students identify tasks like generating ideas or structuring content as areas in which AI can help. They mention limitations of AI like lack of personal voice and lack of emotion. They begin to consider ways in which AI can be a collaborator in creative endeavors.

 **Do This:** Place students in groups of four.

 **Distribute:** Give each student a copy of the Collaborate on a Product Launch activity guide.

 **Do This:** Define *co-creating*.

Teaching Tip

Behind the Curtain: AI tools like ChatGPT and image generators don't really create from scratch, in a traditional sense. Instead, they remix ideas they've encountered before as data. That is why the prompt that we use is so important. A carefully created prompt tells the AI model what to remix, how to do it, and in what voice or style.

 **Do This:** Click through the animated slide to introduce the Product Launch activity.

 **Do This:** Direct groups to choose a product or app concept designed to help students in school.

 **Do This:** Direct each group member to choose a role and record their choice in their activity guide. The career choices are: copywriter, graphic designer, video director, sound designer.

Collaborate on a Product Launch (20 minutes)

 **Goal of this activity:** Give students hands-on experience using AI to co-create part of a product launch. Students will work through a sequence of AI-powered levels in order to build and refine their content.

Remarks

You've seen how creators break tasks into smaller steps. Now you'll use AI to help complete your part of the product launch, just like a real creative team might. Move through the four stages—research, brainstorming, content creation, and refinement.

 **Transition:** Direct students to Lesson 9, Level 1. Students should complete each choice level based on the part of the product they are launching.

 1-5

Collaborate on a Product Launch



 **Circulate:** Check that students are progressing through each stage. Listen for how they talk about AI responses and how they're improving or combining ideas. Encourage specificity in their reflection and remind them that they are the creative lead and that the AI is here to support, not replace, their ideas.

Remarks

Now that you've completed your part of the campaign with AI, let's reflect on the collaboration. Think about what you gained from using AI and what could have been lost if you relied only on it. These questions help us understand AI's strengths and limitations in creative work.

 **Do This:** Direct students to Lesson 9, Level 6 to complete the formative assessment.

 6

Check Your Understanding

Teaching Tip

If students only focus on the benefits of AI (e.g., saving time, generating ideas), prompt them with: "What would have been lost if AI had handled everything on its own? What creative elements did you contribute that AI couldn't replicate?" This encourages them to think about AI as a complement, not a replacement, for human creativity.

Assessment Opportunity

This assessment will help you gauge how well students understand the role of AI as a collaborator. Pay attention to the following:

- **Balanced responses:** If students acknowledge both the benefits of AI and the essential human contributions, they understand how AI can enhance creativity without replacing it.
- **Specific examples:** Are students providing detailed answers based on their roles in the campaign (e.g., how AI helped generate slogans or ideas, and what human creativity brought to the project)?
- **Critical thinking:** Are students able to identify specific aspects of their work that AI couldn't handle, such as adding emotional depth, personal perspective, or fine-tuning content?

Actions to take:

- If students focus too much on AI's benefits (e.g., efficiency, idea generation), remind them of the unique human elements that make creative work impactful, like originality, emotion, and context. Revisit the "AI Essay Critique" activity from earlier in the lesson to highlight these aspects.
- If students struggle to justify their reasoning or give vague answers, prompt them with: "Can you explain how AI's suggestions were useful and where human input was essential?" This will guide students to connect their reflections to the different stages in the "Collaborate on a Product Launch" activity (e.g., brainstorming, refining).
- If students demonstrate strong understanding, celebrate their insights by encouraging them to share with the class. They should be able to explain how AI and human contributions work together to produce a final, polished creative piece.

Next Step:

- If most students are able to reflect critically and balance the strengths and weaknesses of AI, continue with the wrap-up discussion to consolidate their learning.
- If some students need more support, consider having a follow-up conversation or provide extra examples to reinforce AI's complementary role in creative work.

Wrap Up (5 minutes)

 **Goal of this activity:** To encourage students to reflect on how AI enhanced the creativity of their group's product launch project and express their thoughts through discussion.

☒ **Do This:** Review the lesson objectives.

☒ **Do This:** Direct students to "Stand and Swap." First, students stand and discuss the Question of the Day within their original project group. Then, each group "swaps" by joining with another group nearby to share insights. Groups compare experiences and reflect on how AI helped shape or improve their creative process.

☒ **Do This:** Review the Key Vocabulary.



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Lesson 10: Remixing, Creativity, and Originality

45 minutes

Overview

Career Journeys

How do human strengths and AI capabilities interact?

This lesson explores how AI can support human creativity through a hands-on remixing challenge. Students interact with AI to refine both their own creative content and AI-generated work, comparing the role of human input in both processes. The lesson emphasizes that while AI can enhance creative work, human judgment is essential for originality, nuance, and meaning.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- IC - Impacts of Computing

Agenda

Warm Up (10 minutes)

Activity (30 minutes)

Career Exploration

AI Remix Challenge

Synthesis Discussion

Assessment

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Analyze the role of human creativity in AI-assisted content generation.
- Evaluate the strengths and limitations of generative AI.

Preparation

- Prepare copies of the AI Remix Challenge activity guide (one for each student)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **[KEY] Problem Solving with AI Unit Guide Exemplar** - Answer Key

▼ Make a Copy

For the students

- **AI Remix Challenge** - Activity Guide
- **Create a Game Using AI** - Video
- **Problem Solving with AI Unit Guide** - Activity Guide

▼ Make a Copy

Teaching Guide

Warm Up (10 minutes)

 **Goal of this activity:** Have students reflect on AI's ability to be creative by exploring AI-generated beats to reflect different moods.

 **Transition:** Direct students to Lesson 10, Level 1 to experiment with AI-generated drum beats in AI Music Lab. Encourage students to try generating beats for different moods (e.g., elevator music, intense combat scene in a movie, upbeat pop song, etc.) to hear how AI structures patterns.



AI in Music Lab

 **Discuss:** Click through the animated slide to display the prompts.

- *Does this music sound creative? Why or why not?*
- *How does AI decide what beat patterns to generate?*
- *Is AI creating music, or is it following patterns from existing data?*

 **Discussion Goal:** Students should recognize that AI generated music based on patterns and probabilities, not creativity. AI predicts what notes and beats are most likely to follow, but it doesn't compose music with artistic intent like humans do.

Teaching Tip

If students struggle, have them compare AI-generated beats to a song they know. Ask: "What decisions did the human composer make that AI didn't?" or "How does AI's approach differ from an artist's creative process?"

 **Do This:** Explain that AI-generated music is based on recognizing common beat patterns from existing songs. AI predicts what notes and rhythms are likely to follow based on its training data. Unlike humans, AI doesn't compose with emotion or meaning—it just finds the most probable next sound.

Remarks

Now that we've seen how AI generates music by recognizing patterns, let's explore how AI assists in other creative fields like design, storytelling, and game development.

 **Do This:** Introduce the lesson objectives.

 **Do This:** Introduce the Question of the Day.

Activity (30 minutes)

Career Exploration (5 minutes)

 **Goal of this activity:** Help students understand how AI is used in different creative professions.

Remarks

Let's watch a short video to see how professionals use AI in their creative work. As you watch, pay attention to what the human does and what the AI does.

🎥 **Video:** Show the "Create an Entire Game Ad using AI!" video.

Questions to consider with the video:

- What tasks does the human do?
- What tasks does AI assist with?
- Did AI replace human creativity or did it enhance it?

Discussion Goal: Students should identify the different roles that humans and AI play in creative work by analyzing specific tasks performed in the video. The goal is for students to recognize that AI assists with creative processes by generating ideas, refining details, or automating repetitive tasks, but human creativity is still essential for originality, decision-making, and artistic intent.

💡 Teaching Tip

If students struggle to differentiate AI and human tasks, prompt them to think about what decisions required human judgment. Ask: *"What creative choices required human judgment? Could AI have done that task alone?"*

AI Remix Challenge (15 minutes)

🎯 **Goal of this activity:** Apply AI to refine creative content through iteration, evaluation, and human input.

📄 **Distribute:** Give each student a copy of the AI Remix Challenge activity guide.

💡 Teaching Tip

Encourage students to work in pairs so they can discuss AI's suggestions, evaluate improvements, and challenge each other's ideas. Thought partners help reinforce critical thinking and collaboration.

🎤 Remarks

In this activity, you'll collaborate with AI to create and improve a social media post — just like a creative team would. You and the AI will switch off between being the 'pilot' — the one leading the writing — and the 'co-pilot' — the one providing feedback and suggestions. This pilot/co-pilot language comes from pair programming, where two people work together on code — one types, the other watches and guides. Here, you're doing that kind of teamwork, but one of your partners is AI.

☑ **Do This:** Click through the animated slide to help you explain the structure of the activity by walking students through the two rounds of collaboration. In Round 1, students act as the pilot by writing their own social media post, while AI serves as the co-pilot by giving suggestions to revise and improve it. In Round 2, the roles reverse: AI generates a post as the pilot, and students become the co-pilot by reviewing and improving it. Emphasize that both rounds involve thoughtful revision and decision-making.

💡 Teaching Tip

You might pause after explaining both rounds and ask: 'Which role do you think will feel easier? Why?' This primes students to reflect later on which type of collaboration they prefer.

🎤 Remarks

Remember, you are the pilot. Start by choosing one of the social media post prompts in your activity guide. Then write your post on page 2. Keep it short, clear, and school-appropriate — a few sentences is all you need.

Do This: Instruct students to choose one of the social media post prompts in Part 1 of their activity guide. Give them a few minutes to write their initial post on page 2. If students are stuck or short on time, encourage them to use one of the example scenarios provided. Remind students that the goal is to take the lead in content creation and thoughtfully collaborate with AI to strengthen their message.

Teaching Tip

Encourage students to keep their original content short so they have more time to explore AI's suggestions and refine their work.

Transition: Direct students to Lesson 10, Level 2 to work with their AI Chat co-pilot to get suggestions for revision ideas. Make sure students record their revised content in Round 1 of their activity guide.

2

Students as the Pilot

Circulate: Remind students to evaluate the AI's suggestions, not just accept them. Ask: 'Which suggestions do you agree with? Which don't sound right for your audience or tone?' This helps them stay in control of the revision process.

Teaching Tip

Remind students that AI does not replace their creativity—it's a tool to refine their ideas. If time allows, encourage experimenting with multiple AI suggestions to compare which refinements feel the most valuable.

Discuss: Click through the animated slide to display the prompts.

- *Did AI's suggestions improve your content? Why or why not?*
- *Which AI suggestion did you find most useful, and which one did you ignore? Why?*

Discussion Goal: Help students recognize that AI can enhance creative work, but human decision-making is needed to refine originality and intent. Students should consider which AI suggestions actually improved their work and which ones didn't align with their creative instincts. The goal is to get students thinking critically about AI's role as an assistant rather than a creator.

Remarks

Now it's your turn to be the co-pilot. This time, AI Chat will write a social media post as the pilot, and you'll guide it through revising its work — just like it helped you earlier.

Do This: Explain that students are now switching roles. The AI becomes the pilot and takes the lead by writing a social media post based on the same prompt the student used. Students act as the co-pilot — reviewing the AI's draft and offering feedback to improve it. Emphasize that they are helping the AI improve its work, just like it helped them earlier.

Transition: Direct students to Lesson 10, Level 3. Have them prompt the AI to generate a new post based on the same scenario they used earlier. Then, students provide revision suggestions and record both versions in their activity guide.

3-4

Students as the Co-Pilot

3

4

 **Circulate:** As students work, look for students giving vague or overly positive feedback to the AI. Push them to go deeper with questions like, "What exactly would you change?" or "What's missing here?"

Teaching Tip

Some students may find it harder to revise AI content than their own. Encourage them to focus on clarity, tone, audience, and specificity. If a student is stuck, try: "What part of this sounds robotic or repetitive?" or "What's one detail you could add to make this feel more human?"

Synthesis Discussion (5 minutes)

 **Goal of this activity:** Now that students have experienced both roles—getting AI's feedback on their work and guiding AI to refine its content, students compare these two experiences.

 **Discuss:** Click through the animated slide to display the prompts.

- Which type of AI collaboration did you prefer and why? (i.e., getting AI's suggestions on how to refine your original idea vs. guiding AI to refine its content)
- Where would you find each type of collaboration useful in any of the tasks you do both in and outside of school?

 **Discussion Goal:** Students should recognize the difference between AI assisting in content refinement and AI generating original work. The goal is for students to evaluate whether AI's suggestions improved the content or made it feel more artificial, and to reflect on the importance of human input in creative tasks.

Assessment Opportunity

The **activity guide provides evidence** of whether students are critically engaging with AI's suggestions rather than accepting them passively.

The **Synthesis Discussion reveals students' ability** to apply these insights to real-world creative collaboration.

Teachers can determine the next instructional steps based on student understanding:

- **If students mostly relied on AI's suggestions without critically evaluating or modifying them:**
 - Reinforce human decision-making in creative refinement.
 - Revisit AI's role as an assistant by reviewing a sample AI-generated response and asking: "What would make this feel more original?"
 - Have students reattempt refining AI's response, focusing on their own judgment to improve clarity, engagement, and originality.
- **If students struggled to articulate differences between AI-generated creativity and human creativity:**
 - Clarify AI's strengths and weaknesses by asking students: "What was AI able to do instantly? What did you have to refine or adjust yourself?"
 - Facilitate small group discussions comparing AI's role in different tasks (e.g., storytelling, design, music).
 - Use an example of an AI-generated response and discuss what would make it feel more human.
- **If students clearly differentiate AI's role from human creativity and refinement:**
 - Move on to the assessment without reteaching. Students demonstrate an understanding of how AI and human creativity interact and can articulate AI's role as a tool rather than a creator.

 **Do This:** Briefly review each takeaway with students, pausing to reinforce the idea that AI is a tool, not a replacement for human creativity.

 **Remarks**

As we've seen today, AI is great at generating ideas and patterns, but the heart of creativity—meaning, emotion, and originality—comes from you. The most powerful work happens when AI helps you get started and you take it further.

Assessment (5 minutes)

 **Transition:** Direct students to Level 5 to complete an assessment reflecting on their process during the choice level activity. Let students know this assessment is their opportunity to reflect on how AI and human creativity interact.

 5

 Check Your Understanding

 Assessment Opportunity 

Teachers can review **student responses** to assess how well students understood the **role of AI in creative tasks vs. human refinement**.

- If most students describe AI as fully creative (e.g., "AI was the creator" or "AI was the one making the content"):
 - Clarify AI's role as an assistant by reviewing the slides again. Remind students that AI relies on learned patterns to generate responses and does not create with originality or intent like a human.
 - Pose the class a reworded discussion prompt: "How do human strengths and AI capabilities interact?" and have students explain to a partner before moving forward.
- If some students struggle to explain their role vs. AI's role in refinement:
 - Have students pair up and compare answers to discuss what AI contributed vs. what they controlled.
 - Follow up with a Think-Pair-Share on "What creative decisions did YOU make in this process?" before moving to the wrap-up.
- If students clearly differentiate AI's role from human creativity:
 - Move on to the wrap up without reteaching. Students are demonstrating understanding of AI's role in creative tasks.

Wrap Up (5 minutes)

 **Do This:** Review the lesson objectives.

 **Do This:** Have students take out their Problem Solving with AI Unit Guide.

 **Do This:** Explain that they will answer the Question of the Day today by reading the statements in their Unit Guide about AI and creativity. For each one, students should circle a thumbs up if they agree, thumbs down if they disagree, or thumbs sideways if they're unsure. Let students know that they can make themselves notes about their reasons behind what they circled in their Unit Guide if they would like.

 **Do This:** Explain that the next few slides contain the same statements about AI and creativity that they just saw in their Unit Guide. For each one, students should hold up a thumbs up if they agree, thumbs down if they disagree, or thumbs sideways if they're unsure. After each vote, call on one or two students from different positions (e.g., one thumbs up, one thumbs down, one thumbs sideways) to explain their thinking.

 **Do This:** Display statement #1: AI is just as creative as humans. Ask students to respond with a thumbs up, down, or sideways. Then call on one student who agreed and one who disagreed to explain their thinking. Invite a third student if someone gave a sideways response to explain what made them unsure.

💡 Teaching Tip ▲

If time allows, have students explain their choice to a partner first before opening the discussion to the whole class. This helps more students clarify their thinking and increases participation in the full-class share-out.

☑ **Do This:** Display statement #2: I would use AI again for a creative project. Ask students to respond with a thumbs up, down, or sideways. Then invite one student from each category to briefly explain their reasoning. Encourage students to reflect on what worked well (or didn't) when using AI in this context.

💡 Teaching Tip ▲

If time allows, have students explain their choice to a partner first. This gives everyone a chance to process and prepare their reasoning before the whole-class share-out.

☑ **Do This:** Display statement #3: AI and humans working together create better results than AI or humans working alone. Ask students to vote using thumbs up, down, or sideways. Then invite one student from each group to share their reasoning. Emphasize the importance of backing up their opinion with an example from the remix activity or a real-life scenario (e.g., AI in art, music, writing, etc.).

💡 Teaching Tip ▲

If time allows, have students explain their choice to a partner first. This gives everyone a chance to think through their answer before sharing aloud. Encourage students to reference their remix experience or the final reflection to support their reasoning.



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Lesson 11: Ownership, Ethics, and Creativity

45 minutes

Overview

What are the ethical considerations of AI collaboration?

Students explore the ethical tradeoffs of using AI in creative contexts by engaging with real-world controversies involving ownership, attribution, and compensation. Through a values-based warm-up, rotating discussion stations, and a reflection on guiding principles, students consider how AI-generated content affects creators and what fairness looks like in these scenarios. The lesson helps students evaluate AI's role as a collaborator and justify when its use may or may not be appropriate.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- IC - Impacts of Computing

Agenda

Warm Up (5 minutes)

The Reality Check: Real and Fair or Misleading and Unfair?

Activity (35 minutes)

Vocabulary and Real World Examples

Ownership and AI Stations

Check Your Understanding

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Evaluate the role of AI as a collaborator in creative work, including how it can support or undermine the contributions of human creators.
- Justify how AI should be used in creative situations, considering ethical, practical, and societal values.

Preparation

- Review the Ownership and AI Stations Resource to see how to set up your room for the Stations activity in the lesson
- Print copies of the Station Instructions from the Ownership and AI Stations resource.
- Print copies of the Ownership and AI Stations activity guide (one for each student)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- Ownership and AI Stations - Resource [▼ Make a Copy](#)
- [KEY] Problem Solving with AI Unit Guide Exemplar - Answer Key [▼ Make a Copy](#)

For the students

- Ownership and AI Stations - Activity Guide [▼ Make a Copy](#)

Vocabulary

- **attribution** - the act of giving credit to the original creator of a work, especially in the context of using or referencing their material
- **copyright** - the legal right granted to the creator of an original work to use, distribute, and modify it
- **ethics** - a guide for responsible choices about emerging technologies, ensuring privacy, consent, and safety for everyone
- **transparency** - the practice of openly disclosing how something works, especially in terms of decision-making, data use, or AI processes

Teaching Guide

Warm Up (5 minutes)

The Reality Check: Real and Fair or Misleading and Unfair?

 **Goal of this activity:** To spark students' thinking about how AI-generated content can raise questions of fairness, originality, and trust — especially when it blurs the line between human and machine-made work.

 **Do This:** Introduce the warm-up by explaining that students will see real examples of how AI is used to create things like songs, videos, or writing. For each example, give students approximately 30 seconds to decide whether it feels real and fair or misleading and unfair. Then, have them show a thumbs up if it feels real and fair, a thumbs down if it feels misleading and unfair, or a sideways thumb if they're unsure or in the middle. Let students know you'll call on a few volunteers after each round to briefly share why they chose their position.

Teaching Tip

This warm-up is meant to spark curiosity and preview the ethical challenges they'll explore more deeply in the main activity — not to settle any debates yet. Keep the pace brisk: just 30 seconds for students to decide, then a quick reason from 1–2 volunteers. Don't let discussion go too deep here — remind students they'll have more time to dig into these ideas during the station activity.

 **Do This:** Read the scenario aloud: "AI-generated news article." Give students about 30 seconds to decide whether this example feels real and fair or misleading and unfair. Then, them show a thumb up, down or sideways. Call on a few students to briefly explain their reasoning.

 **Discuss:** *Thumbs up if you think an AI-generated news article feels real and fair. Thumbs down if you think it feels misleading and unfair. Thumbs in the middle if you are unsure.*

 **Discussion Goal:** Students begin to question the trustworthiness and authorship of AI-generated content. This example invites discussion about whether AI-generated writing should be trusted, who's accountable for errors, and whether it's ethical to present AI-written news without clear labeling.

 Teaching Tip

You might mention the real-world case of CNET, which published dozens of AI-written articles in 2023. Many contained factual errors or plagiarism and led to public backlash. Ask students, "If the article looks professional but it is wrong, who should be held responsible?"

 **Do This:** Read the scenario aloud: "AI-generated audio recording based on someone's actual voice." Give students about 30 seconds to decide whether this example feels real and fair or misleading and unfair. Then, they show a thumb up, down or sideways. Call on a few students to briefly explain their reasoning.

 **Discuss:** *Thumbs up if you think an AI-generated audio recording based on someone's actual voice feels real and fair. Thumbs down if you think it feels misleading and unfair. Thumbs in the middle if you are unsure.*

 **Discussion Goal:** Help students recognize that AI-generated voice recordings raise serious questions about consent, impersonation, and control over identity. They should begin to consider whether it's ethical to mimic someone's voice — especially when used in entertainment or media — without permission.

 Teaching Tip

Consider highlighting the example of the 2023, AI-generated song with the voices of Drake and The Weeknd going viral even though neither artist was involved.

 **Do This:** Read the scenario aloud: "AI filter or airbrushed photo." Give students about 30 seconds to decide whether this example feels real and fair or misleading and unfair. Then, they show a thumb up, down or sideways. Call on a few students to briefly explain their reasoning.

 **Discuss:** *Thumbs up if you think an AI filter or airbrushed photo feels real and fair. Thumbs down if you think it feels misleading and unfair. Thumbs in the middle if you are unsure.*

 **Discussion Goal:** Encourage students to think critically about how AI tools for filtering or retouching photos influence our perceptions of reality, beauty, and authenticity. They might surface ideas about consent (when someone's image is altered), self-representation, or pressure to meet unrealistic standards. You can also connect this to broader questions of what counts as original or truthful in creative work.

 **Discuss:** Click through the animated slide to display the prompts.

- *If AI content (like a deepfake video or audio) looks or sounds like a real person, but a real person didn't create it, who should get credit or control?*
- *What makes an AI creation feel unfair to the original person or artist involved?*

 **Discussion Goal:** Students start thinking critically about representation and ownership. By referencing deepfakes explicitly, the class can discuss what happens when AI convincingly imitates someone's voice or face — and how that differs from AI remixing someone's creative work. This sets up both the vocabulary (like attribution and copyright) and the upcoming station questions about credit, control, and fairness.

 Remarks

If we can't tell the difference between what AI makes and what people make, how should we handle credit, permission, and payment? That's exactly what we're going to explore in the next activity.

 **Do This:** Introduce the lesson objectives.

 **Do This:** Introduce the Question of the Day.

Activity (35 minutes)

Vocabulary and Real World Examples (10 minutes)

 **Goal of this activity:** Introduce foundational concepts to help students evaluate AI-generated content legally and ethically and analyze AI copyright challenges through real-world disputes.

☒ **Do This:** Have students take out their Problem Solving with AI Unit Guide.

☒ **Do This:** Define *ethics*.

Teaching Tip

After defining ethics, ask students to think about how using someone else's work without asking might be considered an ethical issue — even if it's not illegal. Don't get into examples yet; just note that ethics often involve how people are treated, not just what rules exist.

☒ **Do This:** Define *copyright*.

Teaching Tip

Let students know this is a legal concept, not just about giving credit — it's about who has the right to decide how a piece of work is used. You might find it helpful to tell students that if they create something, they decide whether others can copy it or profit from it. That legal control is what copyright protects.

☒ **Do This:** Define *attribution*.

☒ **Do This:** Define *transparency*.

Teaching Tip

You might tell students, that transparency matters when people don't know if their work is being used, or how AI systems are making decisions. This helps students understand why lack of transparency can lead to ethical problems — especially if creators never know they were involved.

☒ **Do This:** Introduce the idea of transparency policies by explaining OpenAI shares general information about how it trains its models but does not always list exactly what data was used. Encourage students to think about how this may make some creators raise questions about a lack of transparency making it impossible to tell if their work was included without permission and attribution.

Teaching Tip

This is a good opportunity to reinforce the vocabulary words "transparency" and "attribution." Consider asking students how they would feel if their own artwork or writing were used to train an AI without them knowing. However, don't spend too much time on these next three slides as the bulk of this activity should be devoted to student work time.

☒ **Do This:** Introduce the EU AI Act as an example of how governments are starting to regulate AI. Point out that it requires companies to say if they used copyrighted content and gives creators the option to opt out. Use the pyramid graphic to explain how the Act assigns different rules based on an AI system's risk level, with higher-risk systems requiring stronger ethical protections.

☒ **Do This:** Emphasize that this example was one of the strongest reactions to AI training practices and provide the context that in 2023 artists and authors sued AI companies for using their copyrighted works without their permission. Encourage students to consider that this is not just a legal issue but an ethical one as well as it affects

the livelihoods of these creators.

Teaching Tip

Behind the Curtain: A lot of the knowledge from AI models comes from "scraping" the internet (*the act of crawling millions of webpages and analyzing the content contained on them*). As such, the creators of these AI models don't often know where any particular data came from or who it might belong to. This is one reason why it is hard to give credit or to ask permission after the fact.

Ownership and AI Stations (20 minutes)

Goal of this activity: Students analyze real-world AI controversies to explore the ethical tradeoffs between innovation and creative ownership. They'll consider issues like consent, credit, and compensation while hearing multiple perspectives and forming their own opinions.

Remarks

Now that we've looked at some real-world examples of how AI is used in creative work, let's explore the bigger question: How can we balance the benefits of AI with fairness to original creators? At each station, you'll see a real controversy and weigh in with your own perspective.

Do This: Place students in groups of 2-3.

Distribute: Give each student a copy of the Ownership and AI Stations activity guide.

Do This: Explain the Ownership and AI Stations and the steps for this activity. Direct students to complete the stations with their groups. They will have about 2-3 minutes per station.

Circulate: As students work, listen for nuanced reasoning and respectful disagreements. Encourage them to use terms like consent, credit, copyright, and attribution in their discussions.

Do This: Direct students to go back to the station they started at for about 1 minute to take a look at the sticky placement and read their peer's sticky note comments. Once time is up, direct students to the station they went to second and repeat. Remind students to write down any new ideas or reflections they have on their activity guide as they visit each station.

Teaching Tip

Timing: Before students rotate, give them a 10-second countdown to finish writing and posting. After students complete all stations, allow them to revisit their first and second stations to see new responses. (They don't need to revisit their third station — they just came from it!)

Use the GIF: While going through the instructions for the stations, it may be helpful to either talk through the GIF on the slide or demonstrate the flow to ensure students understand that they will be writing on and putting two sticky notes on each poster.

- The first sticky note at each station will be used to place on a scale or on a chart after they write their reasons for their placement on that first sticky note
- The second sticky note at each station will be used to write their answer to the second question and they will place this second sticky note under the question on the poster.

You can determine whether you would like students to put their names on their sticky notes to track participation. If you decide to have students add their names, it is suggested that they put their names on the back of the sticky notes so that students are more focused on the responses, rather than who wrote what.

Do This: Direct students' attention to Station A.

 **Discuss:** Have a quick discussion about the placement of sticky notes on the different quadrants. What do they notice? What do they wonder? Click through the animated slide to display the second question, and pick out a couple of sticky notes from the second question on the poster to help drive the discussion of ownership and copyright.

 **Discussion Goal:** The primary goal of this discussion is for students to explore how AI-generated content raises questions about ownership and copyright. They should consider who should control or benefit from content created by AI — especially when it is trained on human-created works. Students may mention shared ownership, consent, and whether companies or users should have legal rights to something built on other people's creative work.

 Teaching Tip

To keep the station debriefs tight, try to spend about 1-2 minutes per station's poster and focus the discussions on the second question.

☒ **Do This:** Direct students' attention to Station B.

 **Discuss:** Have a quick discussion about the placement of sticky notes along the real to not real scale. What do they notice? What do they wonder? Click through the animated slide to display the second question, and pick out a couple of sticky notes from the second question on the poster to help drive the discussion of authorship and creativity.

 **Discussion Goal:** The primary goal of this discussion is for students to reflect on how AI-generated art challenges definitions of authorship and creativity. They should consider attribution — who deserves credit — when a person provides a prompt, but the machine generates the final output. They may connect this to whether the AI itself, the programmer, or the user should be seen as the "creator," and whether competitions should require disclosure or separate AI-assisted work from human work.

☒ **Do This:** Direct students' attention to Station C.

 **Discuss:** Have a quick discussion about the placement of sticky notes on the Yes or No table. What do they notice? What do they wonder? Click through the animated slide to display the second question, and pick out a couple of sticky notes from the second question on the poster to help drive the discussion of compensation, consent, and transparency.

 **Discussion Goal:** The primary goal of this discussion is for students to analyze how AI-generated music raises concerns about compensation, consent, and transparency. They should explore whether musicians should have control over how their work is used to train AI, and who deserves to be paid when that training leads to profitable outputs. Students may debate whether using someone's style or voice without permission violates ethical or legal standards, even if the AI output is new.

 1

Ownership, Ethics, and Creativity

Check Your Understanding (5 minutes)

 **Goal of this activity:** To give students a chance to reflect on the ethical tradeoffs discussed at each station and articulate a value they think should guide how AI interacts with human-created content.

 **Transition:** Navigate to Lesson 11, Level 1 on Code Studio in order to answer the free response reflection question.

 2

Check Your Understanding

Use this self-assessment to gauge how well students understand the ethical tradeoffs involved in AI development. As students respond, look for whether they:

- Identify a clear value (e.g., attribution, transparency, consent) that reflects an ethical concern
- Reference or imply key terms like attribution, transparency, compensation, or ownership
- Demonstrate awareness of both benefits and risks of using human-created work in AI systems
- Show consideration for who benefits and who may be left out when AI is trained on others' work If many responses are vague or one-sided (e.g., "AI should always be allowed to use whatever data it wants"), consider doing one or more of the following:
- Revisit one of the case studies from earlier in the lesson and discuss: "What might this creator say about that value?"
- Have students revisit the vocabulary terms and identify which ones apply to their response
- Use a class discussion to explore a range of student responses and ask students to challenge or build on one another's reasoning

Wrap Up (5 minutes)

☑ **Do This:** Review the lesson objectives.

🎯 **Goal of this activity:** Students synthesize their learning and reflect on the ethical considerations of AI collaboration.

☑ **Do This:** Have students answer the Question of the Day in their Unit Guide and then share with their peers using a Think-Write-Share protocol:

- Students list ethical considerations in their activity guides
- Students share at one ethical consideration with a partner or small group.
- A few students share insights with the class.

☑ **Do This:** Review the Key Vocabulary.

☑ **Do This:** Review the Key Vocabulary.



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Lesson 12: AI's Role in Society

45 minutes

Overview

How should AI be used in solving real-world problems?

Students choose a real-world example of AI in use within the community, investigating who might benefit and who might be harmed from its implementation. They use their research to draft a call to action in which they propose solutions to the perceived issues that they uncover.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- IC - Impacts of Computing

Agenda

Warm Up (10 minutes)

Activity (30 minutes)

AI in My Community: Helpful or Harmful?

Share Your Voice

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Assess the environmental costs of AI, including energy consumption and carbon emissions.
- Evaluate AI's societal impact by investigating its effects on different community groups.
- Propose equitable AI solutions to ensure responsible and inclusive implementation.

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **[KEY] Problem Solving with AI Unit Guide Exemplar** - Answer Key
▼ Make a Copy

For the students

- **AI In Grocery Stores - 1** - Resource
- **AI in the Community** - Activity Guide
▼ Make a Copy
- **Problem Solving with AI Unit Guide** - Activity Guide
▼ Make a Copy
- **UNICEF Youth Advocate Ayshka on Mental Health** - Video

Teaching Guide

Warm Up (10 minutes)

🎯 **Goal of this activity:** Have students practice weighing pros and cons of an AI-powered system.

 **Do This:** Introduce the scenario about *Newsclues.com*, a fictional website that is deciding to replace the people who pick the articles to feature on the site with an AI-powered news filtering system.

 **Discuss:** Have students write a one-minute response to the discussion prompts about the scenario from the previous slide. Then, have them pair up and take turns summarizing their writing in one minute.

- *What are the risks of using an AI-system to decide which news articles to feature?*
- *Who is impacted most by AI errors, such as linking to unverified or inaccurate sources?*
- *Who is impacted most by AI bias, such as linking to articles that present only one perspective on an issue?*

 **Discussion Goal:** The goal of this discussion is to help students critically evaluate the benefits and risks of AI decision-making in media curation. This discussion will set the stage for deeper exploration of AI's role in their own communities. Here are things to listen for:

- AI is not neutral; it reflects biases in training data.
- AI may favor certain perspectives while excluding others.
- Risk of misinformation (e.g., promoting unreliable sources).
- AI may prioritize engagement over accuracy.
- Benefits: Companies (cost savings), increased efficiency.
- Harms: Journalists (job loss), misinformation in communities.

 Teaching Tip

Pacing: Keep these discussions moving along and once you hear the points mentioned in the Discussion Goal, proceed to the next prompt or section of the lesson. If students have more to share than time allows, let them know that they will have a chance to share their voice later in the lesson as part of the main activity where they write an elevator pitch about AI use.

 **Do This:** Introduce the lesson objectives.

 **Do This:** Introduce the Question of the Day.

Activity (30 minutes)

AI in My Community: Helpful or Harmful? (15 minutes)

 **Goal of this activity:** Help students assess how AI is impacting their own local communities. AI's impact varies by location, economy, and social factors—students should advocate for fair and accessible AI. This part of the activity also focuses on having students use AI to help summarize and draw key points out of source material.

 **Do This:** Place students in pairs.

 **Distribute:** Give each student a copy of the AI in My Community activity guide.

 **Do This:** Direct students to Lesson 12, Level 1. Students choose one implementation of AI to focus on. For all of the choices, students will be sent to a folder with four articles – one feature article about the use of AI they picked, one editorial on the use of AI they picked, one tabloid-style article about the area they picked, and one feature article focused on environmental issues.

 1

Articles

 Teaching Tip

Note: These articles were written by Code.org staff members. The articles were crafted along with ChatGPT, which was used to synthesizing around 20 different news articles on the respective subjects. The facts presented in the articles all come from one or more of these news sources.

You may choose to tailor this list to issues you know are relevant in your students community if you have sources they can use to learn about these issues.

 **Do This:** Direct students to Lesson 12, Level 2. Students choose the same implementation they picked in the previous level and then use AI Chat to summarize the articles and pull out key information. They should record their findings in their activity guide.



2

AI in the Community

 **Circulate:** Listen for students making local connections to their own school or city and asking follow-up questions to the AI when summaries are incomplete. If students finish too fast or seem stuck:

- "How would someone in your neighborhood feel about this?"
- "How could this tech be made more fair or eco-friendly?"

Remarks

You've explored how AI is affecting different parts of your community—now let's shift into reflecting and connecting ideas. Think about the biggest impacts you noticed, any concerns that stood out to you, and what questions you still have about AI's role in society.

 **Do This:** Direct students to Lesson 12, Level 3 to complete the formative assessment.



3

Check Your Understanding

Assessment Opportunity

This reflection will help you gauge how well students understand AI's impact on communities and the environment, and whether they're thinking critically about equity. Pay attention to the following:

- Community awareness: Are students identifying both positive and negative impacts of AI on different groups in their community (e.g., age, income, ability, race, location)?
- Environmental understanding: Do students recognize specific environmental concerns like AI's energy use, reliance on data centers, or carbon emissions?
- Curiosity and gaps: Are student questions thoughtful and relevant, revealing areas where they want to dig deeper—or misunderstandings that need clarification?

Actions to take:

- If students focus too narrowly (e.g., one group or only benefits), prompt them with: "How might this affect someone with different needs or background?" or pair students to challenge each other's perspectives.
- If students skip environmental concerns, briefly review how AI depends on large-scale computing and power use. Encourage connections to articles or earlier class discussions.
- If student questions reveal confusion or patterns of misunderstanding, group similar ones and address them in a brief discussion or mini-lesson to clarify before moving on.

Do This: Tell students to turn to Part 2 of their activity guide. Explain that this is their chance to reflect on everything they've learned so far and come up with one change or improvement that could make AI more fair, more effective, or less harmful — especially for people in their community. Emphasize that it can be a small, practical idea or a bigger, more visionary one. If students are stuck, encourage them to think about who benefits or is left out in their topic, and what could be done differently to improve that.

Circulate: As students brainstorm an idea, encourage them to focus on the core problem or concern they uncovered in their research. Ask questions like: What's something you think should be done differently with this technology?, Who is being left out or harmed in the way this AI is currently used?, or What would make this system more fair or useful for your school or neighborhood? Support students in thinking practically and specifically — their ideas don't need to be perfect or large-scale, just thoughtful and connected to what they learned.

Share Your Voice (15 minutes)

Remarks

We've spent time learning about how AI affects different parts of our community — and you've started thinking about a change you'd like to see. Now we're going to watch a short video from a youth advocate who shares their message about an issue that matters to them. As you watch, see if you can identify the parts of their message that make it powerful — especially the hook, the main idea, and the call to action. These are the same pieces you'll use in your own elevator pitch.

Teaching Tip

Let students know this example is a little longer and more reflective than a typical elevator pitch. For their own pitch, they should aim to keep it short, clear, and direct — something they could realistically say in 30 seconds or less.

Video: Show the "UNICEF Youth Advocate Ayshka on Mental Health" video.

Questions to consider with the video:

- Can you identify the different parts of the pitch — like the hook, the main idea, and the call to action?
- What makes the message feel personal and effective?

Discussion Goal: Students begin to recognize the key elements of an effective elevator pitch — including a strong hook, a clear main message, and a compelling call to action. They also notice how personal experience, emotion, and audience awareness can make a short message more powerful and memorable.

Do This: Explain that students are now moving on to Part 3 of their activity guide, where they'll turn their ideas into a clear message they can share with others. First, they'll choose an audience that would benefit from or care about what they've learned. Then, they'll write a short elevator pitch using the guide on page 3. Students can collaborate with AI — by revising their own draft with AI's help or improving a draft written by AI. Encourage students to think about what kind of message would connect with their audience — should it sound formal or casual? Informative or persuasive?

Circulate: As students work, look for them making connections between their pitch and what they learned in Part 1. If they're unsure how to start, remind them they can begin with the hook and write one sentence at a time. Prompt them to use vocabulary like "equity," "access," "impact," or "responsibility" if relevant.

Transition: Let students know that if they want help drafting or refining their elevator pitch, they can go to Level 4 to use AI Chat. They can choose to be the pilot (write it themselves and ask AI to revise) or the co-pilot (ask AI to draft and then improve it). Using AI is optional — students can also write and revise on their own if they prefer.

Teaching Tip

Some students might benefit from using AI to help them with tone or structure, especially if they're stuck. Others may prefer to write on their own. Either approach is okay — the goal is to clearly communicate their idea to the audience they chose.

☑ **Do This:** Explain that students will participate in a "Give One, Get One" elevator pitch exchange. They will find a partner and take turns sharing their elevator pitches — one person gives their pitch, the other listens and gives quick feedback or asks a question. Then they switch roles. After both students have shared, they'll find a new partner and repeat. Encourage students to share with multiple classmates to hear a variety of perspectives and practice tailoring their pitch to different audiences.

☑ **Do This:** Play the music clip to signal the start of the activity. Use the same music cue consistently so students recognize when it's time to begin "Give One, Get One."

🔄 **Circulate:** As students share their elevator pitches, listen for how clearly they're communicating their main idea and call to action. Offer support to students who may be stuck or rushing through their pitch. Prompt partners with questions like:

- "What did you find most interesting about your partner's idea?"
- "Was anything unclear or missing?"
- "Did their pitch include a strong hook and a clear call to action?"

Teaching Tip

Give One, Get One encourages students to seek and share ideas and information. When prompted by the teacher, students circulate the classroom and pair up with a partner. Each student "gives" or shares one of their ideas as the other student "gets" or listens and writes it down. The teacher then prompts students to find new partners and repeat the process.

Using music to cue transitions helps create routines in the classroom. When students hear cues, they know it is time to transition to a specific activity. Emphasize to students what the music clip means and be consistent with using the music clip to cue the activity. **Click the blue play icon on the slide to play the music cue.**

Wrap Up (5 minutes)

☑ **Do This:** Review the lesson objectives.

☑ **Do This:** Have students take out their Problem Solving with AI Unit Guide.

☑ **Do This:** Review the Question of the Day.

☑ **Do This:** Have students form small groups of three or four.

☑ **Do This:** Explain that now that they have reflected in their Unit Guide, they will share key takeaways with their group. Explain that the "Pass the Mic" discussion is designed so that everyone gets a chance to share. Each person will share their answer to the Question of the Day in one or two sentences and then "pass the mic" so that everyone has a turn.

🔄 **Circulate:** As students share in their groups:

- Listen for common themes and misconceptions such as:
 - Are students clearly identifying AI's strengths and weaknesses?
 - Are students misunderstanding AI's capabilities (e.g., assuming AI "thinks" like a human)?
- Prompt deeper thinking with guiding questions such as:
 - Did AI ever surprise you with what it could or couldn't do?
 - Would you trust AI more or less after today's lesson?

- Did anyone in your group have a different perspective that made you think differently?

💡 Teaching Tip ▲

If time allows, facilitate a full-class discussion of the QotD. Consider asking students to raise their hands if they heard a new perspective in their group that made them think differently.

- If students only state a general opinion (e.g., "AI is good at math"), follow up with: "What makes AI good at that task? Is it better than a human?" or "Can you give an example of when AI would fail at that same task?"
- If some students are not engaging in discussion, ask them directly: "What did you write down in your Unit Guide? Can you share one thing that stood out to you?" or "Did you hear something from your group that you agree or disagree with?"



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Lesson 13: AI Time Capsule Project

135 minutes

Overview

In this project, students work with AI to develop a future prediction and create a presentation that explains their process. They collaborate with AI to generate and refine ideas, evaluate outputs for accuracy and bias, and reflect on how they shaped the final prediction. The project centers on ethical reasoning, student voice, and thoughtful use of multimodal inputs to support transparency and agency in AI collaboration.

Agenda

Day 1 (45 minutes)

Warm Up

Project Overview

Exploring Future Possibilities with AI

Wrap Up

Day 2 (45 minutes)

Warm Up

Activity

Wrap Up

Day 3 (45 minutes)

Warm Up

Finalizing the AI Time Capsule

Presenting Your Prediction and Peer Feedback

Wrap Up

End of Unit Survey

Objectives

Students will be able to:

- Create a presentation that explains how you collaborated with AI to shape and improve your final prediction.
- Develop a future prediction by combining research with AI-generated insights.
- Evaluate AI outputs for accuracy, inclusivity, and real-world relevance.
- Refine predictions by improving prompts and applying human reasoning.

Preparation

- Print copies of the AI Time Capsule Project Planning Guide (one per student)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the students

- [AI Time Capsule Project Planning Guide](#) - Activity Guide

▼ Make a Copy

Teaching Guide

Day 1 (45 minutes)

Warm Up (5 minutes)

☒ **Do This:** Place students into groups of two or three. Display the mix of real ("AI Will Soon Take Your Order at Taco Bell and Pizza Hut") and AI-generated future news headlines and have students analyze the headlines.

 **Discuss:** Click through the animated slide to display the prompts.

- *Which headlines feel possible? Which feel exaggerated?*
- *How do current trends shape what we believe about the future?*

 **Discussion Goal:** Students should recognize that AI-generated predictions can be plausible but also exaggerated or biased based on the data AI has been trained on. The goal is for students to start questioning AI-generated content and assessing its realism.

Teaching Tip

If students struggle, prompt them to consider what evidence would be needed to make a prediction believable.

Remarks

Now that we've examined some AI-generated headlines, let's shift our focus to see how we're going to create our own predictions with AI's help.

Project Overview (10 minutes)

 **Goal of this activity:** Introduce students to the AI Time Capsule project, its goals, and expected deliverables.

 **Distribute:** Give each student a copy of the AI Time Capsule Project Planning Guide.

☒ **Do This:** Explain that in this project, students will be working with AI to explore possible futures. Introduce the project and explain the project structure:

- Students will create a future prediction using AI.
- They will refine it through AI and peer feedback.
- They will present their final work in a creative format.

☒ **Do This:** Instruct students to review the assessment criteria rubric and final deliverables on the last pages of the Project Planning Guide. Explain the design challenge framing of "How can we use AI as a creative partner to generate meaningful future predictions?"

Teaching Tip

Encourage students to think about how AI can be a useful tool for prediction, but not an absolute truth-teller.

 **Discuss:** Click through the animated slide to display the prompts.

- *What do you notice about the project rubric?*
- *What does success look like for this project?*

 **Discussion Goal:** Students share what they notice about the project requirements and rubric and identify the tasks they will need to complete for this project.

 **Discuss:** *What barriers do you anticipate to achieving success on this project?*

 **Discussion Goal:** Students identify barriers they anticipate and share any questions they have about the project.

☒ **Do This:** Click through the animated slide to introduce the steps students will take on Day 1.

☒ **Do This:** Click through the animated slide to introduce the steps students will take on Day 2.

☒ **Do This:** Click through the animated slide to introduce the steps students will take on Day 3.

Exploring Future Possibilities with AI (20 minutes)

 **Goal of this activity:** Guide students through AI-assisted future predictions, evaluating and refining outputs.

Remarks

Now that you understand the project's structure, let's begin shaping your predictions. We're going to start by gathering information, making an initial prediction, and then seeing how AI builds on it. Your job is to refine AI's ideas into something you truly believe could happen.

☒ **Do This:** Instruct students to select a focus area for their project and record it in Step 1 of their Project Planning Guide.

 **Goal of this activity:** Gather enough artifacts to give AI enough context to make a prediction.

☒ **Do This:** Emphasize to students that they should find several resources but that they don't need to look for the "perfect" artifact. Instead, their focus should be on only spending a few minutes gathering resources so they can move on to the next steps.

 **Circulate:** Support students who might be spending a long time searching by asking "What kind of information do you think AI will need?" or "Do you already have a Few artifacts that feel useful?"

Teaching Tip

Make a plan beforehand for how you want students to store artifacts. In order to upload them to AI Chat, students need to save the files to their computers. You might consider leading a mini-lesson for students on how to create a folder on their computer's desktop where they can store images and PDFs.

☒ **Do This:** Direct students to complete the My Initial Prediction section of their planning guide. They should make a prediction using the artifacts they gathered for context.

 **Transition:** Direct students to Lesson 13, Level 1 to work with AI Chat to generate on a future prediction for their chosen area.

 1

AI Prediction

 **Circulate:** Ensure students are working through the steps in the level to refine their output. Have students working at the same pace answer the discussion questions in Step 2 together.

Teaching Tip

Check if students are evaluating AI's predictions critically or just accepting them. If students accept AI's outputs without questioning, ask:

- *"What trends or real-world examples support this prediction?"*

- "Who might benefit or be excluded from this future?"

Encourage students to refine AI's predictions by comparing the predictions to their own knowledge and research.

Wrap Up (10 minutes)

☒ **Do This:** Direct students to answer the self reflection questions in Day 1 Reflection in their Project Planning Guide. Then have students make a final prediction statement using the sentence starter "I believe . . .".

Teaching Tip

Optionally, you can have students record their answers to these questions on a 3x5 card to be handed in as an Exit Ticket as they leave the class.

Day 2 (45 minutes)

Warm Up (10 minutes)

 **Goal of this activity:** Help students recognize how human reasoning improves AI-generated content. They will apply these skills as create a presentation that documents their own process of working with AI.

 **Discuss:** *What exactly did the human change or add?*

 **Discussion Goal:** Students identify the specific changes the human made to the AI's prediction, such as clarifying limits of automation and including a more realistic role for humans. This will help them later in the lesson explain their own changes to the AI outputs they received.

Remarks

Just like in this example, your job today is to explain the refinements you made while working with AI.

Activity (35 minutes)

☒ **Do This:** Review the tasks for Day 2 of the project and explain that today, students' goal is to begin developing their final product.

Teaching Tip

Decide in advance what platforms students will have available for their final product and where they will save their work. If you plan to have students create a slides presentation, infographic, news article, etc. you may want to create a shared class folder where all students can save their products.

 **Goal of this activity:** Guide students in selecting their final presentation format.

Remarks

Before we start developing our final product, let's decide how you want to present your prediction. Think about what format will best communicate your ideas.

☒ **Do This:** Instruct students to pick a product format for their project.

☑ **Do This:** Direct students to complete the Develop Your Presentation section in their Project Planning Guide.

☑ **Do This:** Review the checklist with students to ensure they understand what they need to include in their project. Do this before they move on to Lesson 13, Level 2 and begin collaborating with AI. Students can also find this list in the Collaborating with AI section of their Project Planning Guide.

🖥 **Transition:** Direct students to Lesson 13, Level 2 to use AI as a collaborator to help them refine and improve the content or design of their presentation.

🖥 2

Final Product

🔄 **Circulate:** Ensure students are updating their presentations with the collaboration of AI Chat.

💡 Teaching Tip ▲

As students are creating their project's product, look for if students are integrating AI-generated insights while maintaining their own voice. If students over-rely on AI's wording, ask:

- *"How can you make this sound more like your perspective?"*
- *"What details could you add to make this clearer for your audience?"*

Guide students to refine AI's content rather than copying it directly.

☑ **Do This:** Direct students to exchange projects with a peer and complete Peer Review. Students will use the peer feedback to make revisions to their final product.

💡 Teaching Tip ▲

If time allows, you can have students go through one more round of iteration on their project's product after getting peer feedback.

Wrap Up (5 minutes)

☑ **Do This:** Instruct students that before they move on to finalizing their projects next class, to take a moment to reflect on today's refinements. Direct students to answer Day 2 Reflection questions in their Project Planning Guides.

💡 Teaching Tip ▲

If time allows, have students share their reflections with a partner or in small groups.

Day 3 (45 minutes)

Warm Up (10 minutes)

🎯 **Goal of this activity:** Encourage students to reflect on the importance of making their own voice, decisions, and reasoning clear in their presentation, rather than simply repeating AI-generated content.

🗣 **Discuss:** *How can you tell if something was written by a person or by AI? Why does that matter in your project?*

💡 **Discussion Goal:** Students should recognize the value of personal input in AI-generated content and identify what makes a presentation feel original, human, and thoughtful. They should begin thinking about how to show ownership over their choices and reasoning.

Finalizing the AI Time Capsule (15 minutes)

☑ **Do This:** Give students about 15 minutes to make any final revisions for their project.

Presenting Your Prediction and Peer Feedback (15 minutes)

☑ **Do This:** Have students form groups of three or four students. Direct students to present their projects and engage in peer discussion. Make sure each presenter is answering the following questions as well as any questions their peers have for them:

- *What perspectives are represented in this prediction?*
- *What biases might have shaped this future scenario?*
- *How does this prediction encourage responsible AI use?*

💡 Teaching Tip

Depending on class size, consider using stations, sticky notes for written feedback, or rotating peer groups to keep discussions efficient.

Look for if students are engaging critically with each other's predictions. If discussions lack depth, ask:

- "What makes this prediction compelling or questionable?"
- "What evidence supports or challenges this scenario?"

Push students to ask thoughtful questions and challenge assumptions.

Wrap Up (5 minutes)

☑ **Do This:** Have students think about AI's role in shaping the future and how we ensure AI is used responsibly. Have students complete the End of Project Reflection in their Project Planning Guide.

💡 Teaching Tip

If time allows, have students share their reflections with the class.

End of Unit Survey

After completing this project, students should complete the End of Unit Survey linked in the final level. Their responses help us understand their experiences and improve future versions of this unit.

📄 3

End of Unit Survey



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